

Regulatory pressure and fire sales in the corporate bond market[☆]Andrew Ellul^a, Chotibhak Jotikasthira^b, Christian T. Lundblad^{b,*}^a Indiana University, Bloomington, United States^b University of North Carolina, Chapel Hill, United States

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ABSTRACT

This paper investigates fire sales of downgraded corporate bonds induced by regulatory constraints imposed on insurance companies. As insurance companies hold over one-third of investment-grade corporate bonds, the *collective* need to divest downgraded issues may be limited by a scarcity of counterparties. Using insurance company transaction data, we find that insurance companies that are relatively *more constrained* by regulation are more likely to sell downgraded bonds. Bonds subject to a high probability of regulatory-induced selling exhibit price declines and subsequent reversals. These price effects appear larger during periods when the insurance industry is relatively distressed and other potential buyers' capital is scarce.

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1. Introduction

This paper investigates fire sales of downgraded corporate bonds induced by regulatory constraints imposed on insurance companies. Shleifer and Vishny (1992) show

that forced selling of industry-specific assets may yield transaction prices that are significantly below fundamental values. In these circumstances, other buyers in the same industry may also face financial constraints and therefore be unable to provide liquidity. The presence of

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potential buyers is a crucial factor in determining the price at which any transaction will take place. As Pulvino (1998) argues, the maximum price that potential buyers are willing and able to pay for the asset depends on their own valuation and funding. As an interesting parallel among financial assets, the trading activity of bonds downgraded to speculative-grade by regulatory-constrained insurance companies provides an ideal environment in which to study forced selling and any associated price effects. As a group, insurance companies hold over one-third of outstanding investment-grade bonds (see Schultz, 2001) and, at the same time, face regulations that either prohibit or impose large capital requirements on the holdings of speculative-grade bonds.¹

The empirical literature on asset fire sales provides several situations in which transaction prices may deviate from fundamental values.² Coval and Stafford (2007) study equity market transactions induced by open-end mutual fund redemptions. Mitchell, Pedersen, and Pulvino (2007) investigate the price reaction of convertible bonds around hedge fund redemptions. Pulvino (1998) studies commercial aircraft transactions initiated by (capital) constrained versus unconstrained airlines. Finally, Campbell, Giglio, and Pathak (2009) consider forced selling in the real estate market due to events such as foreclosures. In contrast, we examine a new channel that may induce fire sales: regulatory constraints. Specifically, regulations governing the investing behavior of insurance companies may effectively force the sale of certain assets (particularly those reclassified into a higher risk category) and simultaneously prevent other insurance companies from stepping in as buyers.³ As such, forced selling is most likely to occur in the downgraded bonds that are held by regulatory-constrained insurers such as those that have low risk-based capital ratios. Further, when coupled with a relative lack of liquidity, forced selling is likely to generate significant price pressure.

To empirically test the fire sale hypothesis in the corporate bond market as related to regulatory requirements, we construct a data set of 1,179 bonds that were downgraded to speculative-grade over the period 2001–2005. We combine information on these bonds with observations on individual insurance companies' holdings and transactions provided by the National Association of Insurance Commissioners (NAIC). In addition, we obtain data on the financial position and strength of each

insurance company from the Street.com. We employ an industry-standard risk-based measure to determine which individual insurance companies have the lowest risk capacity and are thus most likely (to have) to sell the downgraded corporate bond if the regulatory pressure hypothesis is valid. In studying the determinants of price patterns around corporate bond downgrades, we also use these constraint measures to make a distinction between trading volume arising from forced selling versus that which is discretionary. Finally, we use the overall distress in the insurance industry and the capital of potential buyers of speculative-grade bonds as time-conditioning variables to examine whether fire sales take place mostly in periods where sellers are under stress and buyers are hard to find.

Several key empirical results deserve attention. First, we find that relatively constrained insurance companies are more likely to immediately sell (at least part of) their holdings of a downgraded corporate bond. Observable selling pressure varies with measurable differences in the financial health of the insurance companies holding these bonds. This result obtains even after controlling for insurance company and bond characteristics (e.g., such as the bonds' general level of liquidity). Second, we find that forced selling of downgraded bonds held disproportionately by regulatory-constrained insurance companies causes prices of these bonds to fall below fundamental values for several weeks around the downgrade event. Downgraded bonds largely held by relatively unconstrained insurance companies do not exhibit these dramatic price effects. Indeed, the median cumulative abnormal return differential across the two groups of bonds separated by the relative constraints of the bond holders is around –6% to –7%. As we observe that the relatively constrained firms have a larger probability of selling downgraded bonds, the evidence again points to a market imbalance generated by regulatory pressure. Further, prices fully revert by week +35, suggesting the imbalance is temporary. These are the key results of the paper.

Why do these patterns exist and where are the potential buyers? Duffie, Garleanu, and Pedersen (2007) provide some recent theoretical guidance. They demonstrate that a market-wide liquidity shock that forces a large group of investors to sell a security, coupled with high search costs, can result in an extended period over which prices deviate from fundamental values. The speed with which transaction prices recover depends to a large extent on counterparty search costs and the associated level of market liquidity.⁴ In illiquid environments, a price recovery may take a significant period of time as market participants await sufficient counterparties. Since insurance companies face shared regulation, the resulting collective need to divest downgraded issues may be limited by a scarcity of counterparties and associated bargaining power. Fire sale prices will obtain, leading to

¹ Other institutions, such as pension funds, also face limitations in holding non-investment-grade bonds. This severely limits the set of potential buyers.

² Boyson, Helwege, and Jindra (2010) show that during crises, a majority of commercial banks, investment banks, and hedge funds use a variety of alternatives to avoid selling affected assets at fire sale prices.

³ Insurance companies are regulated at the state level. Many states adopted the guideline prescribed by the National Association of Insurance Commissioners (NAIC). Holdings of non-investment-grade bonds are generally limited by two types of regulations. First, the capital requirements are 4.6% and 10% for the holdings of BB-rated and B-rated bonds, respectively. The same requirement is only 1.3% for BBB-rated bonds and 0.4% for bonds rated A or above. Second, the NAIC prescribes a hard cap of 20% for all non-investment-grade bonds as a percent of the portfolio; however, this is generally not binding. Some states impose even more stringent guidelines.

⁴ Duffie, Garleanu, and Pedersen (2002), Green, Hollifield, and Schurhoff (2007), and others demonstrate empirically that search costs are important in describing trading in over-the-counter markets.

transaction prices significantly below fundamental values. Liquidity provision, then, has to come from outside of the insurance industry. The gradual emergence of alternative investors in high search cost markets, like that for downgraded corporate bonds, will generate a slow reversal of transaction prices and an eventual realignment with fundamental values. In his AFA presidential address, Duffie (2010), provides a survey of recent research on these price effects across multiple markets associated with slow moving capital.

Price pressures for downgraded bonds are expected to be *larger* when the insurance industry as a whole is in distress and/or when potential outside buyers' capital is relatively scarce. To explore this prediction, we measure selling pressure and price effect differentials across time periods. We find that the selling pressure around the bonds' downgrade and the associated price effects are more pronounced when (a) the insurance industry as a whole is in distress and (b) the capital of potential outside buyers for the downgraded bonds – such as high-yield mutual funds and high-yield and distress-focused hedge funds – is relatively scarce.

Clearly, one important caveat is the fact that the fundamental value is not directly observable. Hence, measuring departures from fundamental values requires some explanation. Following Coval and Stafford (2007), we identify the fundamental value *ex post* by documenting the systematic patterns of transaction prices over time. Evaluating fundamental values in this way is admittedly complicated by the fact that the price effects that we find could be due to two alternative factors beyond the fire sales hypothesis.

The first alternative is the information content of the credit rating change. To address this issue, we employ controls for the information contained in the downgrade by using bond-issuer stock returns around the event. Second, we also consider a sample of bonds issued by firms whose stock returns exhibit no significant reaction to the rating change in the spirit of Ambrose, Cai, and Helwege (2009). Arguing that these bond transactions are “likely to represent regulatory pressure rather than information-motivated trading,” they provide compelling evidence that price pressure is non-existent among this important subset of bonds. While this would appear to stand in contrast to our findings, the major difference between our paper and Ambrose, Cai, and Helwege (2009) is our principal focus on the cross-section and time-series as it relates to the financial health and regulatory constraints of the insurance companies holding the downgraded bonds. It is only because of the heterogeneity afforded by our use of individual insurance company data that we can identify fire sales. For this smaller sample, we continue to observe significant price effects for those bonds largely held by relatively constrained insurance companies; the differential median cumulative abnormal returns between bonds held by more or less regulatory-constrained insurance companies is around -4% in the first five weeks after the downgrade. In line with their conclusions, however, we find no significant price effects for bonds that are held by insurance companies that are not regulatory-constrained.

The second factor relates to pure price pressure offered by Scholes (1972). Under this alternative explanation, transaction prices diverge from fundamental values because of an uninformed shock caused by either excess demand or supply. Providers of liquidity have to be compensated and this results in transaction price overshooting.⁵ Distinguishing between pure price pressures and fire sales is problematic because the two phenomena are not mutually exclusive. Our line of argument is that while price pressures can occur when corporate bonds are downgraded irrespective of the financial health of the investors holding them, fire sales should only occur when the bonds are disproportionately held by regulatory-constrained insurance companies that are forced to sell and potential buyers' capital is scarce, making it difficult to find a counterparty who is willing to pay competitive prices or who values downgraded bonds as much as the sellers. It is this dimension that we want to emphasize in this paper.

The remainder of the paper is organized as follows. Section 2 discusses the sample construction and describes the summary statistics of the data. Section 3 provides results on how regulatory constraints at the insurance company level, insurance industry distress, and the scarcity of buyers' capital influence selling activity and price pressures around the downgrades. Section 4 presents cumulative abnormal returns and trading volume around the bond downgrades. Section 5 investigates several alternative explanations. Section 6 concludes.

2. The data

2.1. Sample construction

We use three main sources of data. First, the insurance companies' transaction and year-end position data are from the National Association of Insurance Commissioners (NAIC). Second, the information on the characteristics of each bond (including maturity, offering date, credit rating, etc.) is from Mergent Fixed Income Securities Database (FISD). Third, the annual information on the financial strength of each insurance company is from the Street.com Ratings (SR). SR is a wholly owned subsidiary of the Street.com, a provider of financial information and company analysis reports. SR provides financial strength ratings, as well as the components that are used to construct the ratings, for financial institutions, stocks, mutual funds, and insurance companies. In addition to these three main sources, we also use time-conditioning and stock-specific variables from a variety of other sources, including Center for Research in Security Prices (CRSP) and Datastream, to either control for fundamental price movements or identify periods over which fire

⁵ The evidence so far shows that such price overshooting is temporary, and can last from a few hours (Gemmell, 1996; Ellul, 2006) to a few days (Harris and Gurel, 1986; Kraus and Stoll, 1972; Scholes, 1972; Shleifer, 1986). On the other hand, the fire sales story is likely to have a much longer temporal impact on prices, with slow reversals towards the fundamental value taking weeks or months.

sales are more likely. Our sample period is from 2001 to 2005.⁶

NAIC data, for different sample periods, are used by Schultz (2001), Campbell and Taksler (2003), Krishnan, Ritchken, and Thomson (2005), and Bessembinder, Maxwell, and Venkataraman (2006), among others. While the NAIC data are not exhaustive, they represent a substantial portion of the corporate bond market.⁷ Schultz (2001) estimates that insurance companies collectively hold between one-third and 40% of investment-grade corporate bonds. Comparing the NAIC data to the more complete Trade Reporting and Compliance Engine (TRACE) data for TRACE-eligible bonds, Bessembinder, Maxwell, and Venkataraman (2006) find that insurance companies completed about 12.5% of dollar trading volume in these bonds during the second half of 2002. In sum, insurance companies represent an important part of the corporate bond market.

NAIC transaction data include detailed transaction information including insurance company identification, bond identification, dealer identification, trade date, direction, price, and size. As Bessembinder, Maxwell, and Venkataraman (2006) point out, NAIC transaction data do not contain transaction time so any estimation that requires time ordering of trades is not readily possible.⁸ However, they show in a simulation that if returns are calculated based on a trade on the prior trading day, the parameter estimates of the Huang and Stoll (1997)-type indicator regression will be unbiased. In the subsequent analysis of abnormal returns around the downgrade events, we follow their suggested method.

NAIC position data provide year-end holding information including insurance company identification, bond identification, holding size in par and market value terms, and other accounting information. We merge the position data with the transaction data to infer quarter-end positions from year-end positions. For each bond in each quarter, we use the latest quarter-end positions (before the event) to identify which insurance companies hold the downgraded bonds and to calculate the aggregate holders' characteristics, including the regulatory constraints they face.

We merge the NAIC transaction and position data to FISD static bond characteristics by bond CUSIP. We use bond characteristics to screen bonds and as control variables. We use a separate rating history file from FISD to identify credit ratings for each bond at each point in time and the rating change events. We use the lower of

Moody's and Standard & Poor's (S&P) ratings. We define the rating change event as the date of downgrade announcement by the rating agency that acts first.

Before proceeding, our downgrade definition requires some explanation since it is not perfectly consistent with that used by NAIC. NAIC considers a bond "speculative-grade" if the bond no longer holds an investment-grade rating. This implies that we should instead define the event date as the date of announcement by the rating agency that last downgrades the bond.⁹

For several reasons, we argue that the de facto downgrade date corresponds to the first downgrade by either Moody's or S&P.¹⁰ First, the initial downgrade to speculative-grade made by either Moody's or S&P is often followed quickly by the other agency. For our sample, 1,149 downgraded bonds are rated by both agencies and 78% of them see both agencies downgrading. Moreover, when both downgrades fall in our sample period, the median number of days between the downgrade made by the first agency and the second agency is just six days. An awareness of the fact that the first downgrade is likely to be soon followed by a second has significant importance for constrained insurance companies holding that bond. Second, prior to July 2005, Lehman Brothers' bond indexes (which are used as investment guidelines by many mutual funds) based their index constituency on the first downgrade of either Moody's or S&P (Chen, Lookman, Schurhoff, and Seppi (2009)). This implies that high-yield mutual funds can buy the downgraded bonds and constrained insurers may want to start selling quickly. Third, insurance companies holding these bonds in limbo may face additional reputational costs with regulators, financial markets, and customers. While measuring reputation is extremely difficult, we know anecdotally that some insurance companies conduct more stringent risk management than is required by NAIC regulation to avoid additional market scrutiny ahead of formal regulatory action. An insurance company may be concerned about alternative market information—for example, virtually all A.M. Best reports on insurance companies touch on investments in high-yield bonds and potential credit losses.¹¹

In the final step, we merge SR insurance company data to our NAIC transaction and position data by NAIC company code. SR uses information on insurance companies from many sources, including the NAIC, to calculate different measures (some are based on prescribed formulas while others are based on proprietary formulas) of financial strength of insurance companies.¹²

⁶ NAIC data that we have cover full years from 2001 to 2005. FISD data end in August 2005; as a result, we exclude transactions on bonds that are issued in or after September 2005. The Street.com data include year-end financial standings from 2000 to 2007.

⁷ We do not use the Trade Reporting and Compliance Engine (TRACE) data, which cover a far greater portion of transactions in the corporate bond market, because TRACE did not cover non-investment-grade bonds until 2005 and did not have customers' buy/sell indicators until November 2008. Also, TRACE does not report identities of traders. Edwards, Harris, and Piwowar (2007) and Goldstein, Hotchkiss, and Sirri (2007) use the non-public version of TRACE.

⁸ This difficulty only applies to the case in which more than one trade takes place on the same day, which is not very common in our data.

⁹ During our sample period, there are three rating agencies (S&P, Moody's, and the smaller Fitch) that are recognized by regulators. Our results hold regardless of whether or not we consider Fitch ratings.

¹⁰ In unreported results, we find that our results based on the first downgrade remain largely unchanged when using the last downgrade.

¹¹ A.M. Best is the most widely recognized agency that rates insurance companies. Institutional investors and customers often use A.M. Best ratings and reports as guidance in deciding whether to do business with a particular insurance company.

¹² NAIC provides a variety of data suites covering different dimensions of insurance companies, extracted from the quarterly and annual reports the companies submitted. Our transaction and position data are from Schedule D.

Table 1

Data screening and merging process.

This table describes how the data are screened and merged. The raw sample is the NAIC transaction data for the period from 2001 to 2005. First, the sample is cleaned and then merged with (i) the rating and FISD bond characteristics and (ii) the Street.com Ratings insurance company data. The downgrade sample is constructed by including only bonds that are downgraded from investment (BBB – and above) to non-investment (BB+ and below) grades during the sample period. The downgrade sample is merged with CRSP's issuer stock data, and the bonds that exhibit insignificant stock return upon the downgrade are identified. The table reports the numbers of transactions, numbers of bond issues, numbers of issuers, and numbers of insurance companies behind the transactions.

	Transactions			Bond issues	Issuers (CUSIP)	Insurance companies
	Life & health	Property	Total			
<i>Total sample</i>						
(1) Raw sample (2001–2005)	1,136,515	571,733	1,708,248	119,790	29,468	3,476
Data errors, non-secondary market trades, and potential outliers ^a	(630,101)	(258,860)	(888,961)	(57,323)	(12,336)	(206)
(2) Merged with credit ratings and FISD bond characteristics	399,470	264,976	664,446	23,276	6,040	3,166
Bonds that (i) are not debentures or medium-term notes (MTN), (ii) have issue size smaller than \$50 million, (iii) have sinking funds, or (iv) are denominated in foreign currencies. Trades that are before offering or after maturity	(111,144)	(65,888)	(177,032)	(9,186)	(1,812)	(82)
(3) Merged with the Street.com data ^b	288,180	195,530	483,710	14,074	4,228	3,042
<i>Downgrade sample</i>						
(4) Cleaned downgrade sample from merged data (3)	40,961	25,060	66,021	1,179	417	2,451
(5) Merged with CRSP stock information ^c	34,038	22,017	56,055	1,043	335	2,404
(5.1) Meeting requirements for abnormal return study ^d	19,692	14,030	33,722	356	173	2,122
(6) Exhibiting zero-issuer-stock returns upon downgrade ^e	20,250	13,133	33,383	584	221	2,144
(6.1) Meeting requirements for abnormal return study ^d	12,627	9,030	21,657	237	117	1,924

^a Examples of non-market counterparties include called, cancelled, conversion, direct, maturity, put, redemption, tax-free exchange, etc. An outlier is defined as a transaction whose product of immediately preceding and subsequent returns is negative and greater than 0.09 in absolute value.

^b The Street.com may not evaluate some insurance companies for any of the following reasons: (i) total assets < \$1 million, (ii) premium income for the current year is < \$100,000, (iii) the company functions almost exclusively as a holding company, or (iv) the Street.com does not have enough information.

^c Each bond is first matched with the issuer stock identifier through the six-digit issuer CUSIP. The unmatched bonds are then matched with the stocks through stock tickers of the bond issuers obtained from Bloomberg and TRACE.

^d For reliability, abnormal return studies use only bonds that trade: (i) at least once before week –20, (ii) at least once between weeks –20 and 0, (iii) at least once between weeks 0 and +35, (iv) at least once after week +35, and (v) at least ten times during the sample period.

^e Following Ambrose, Cai, and Helwege (2009), the zero-issuer-stock return sample is identified by comparing the average daily stock return over the three-day window around the downgrade (days –1 to +1) with the average historical stock return over 250 days (days –280 to –31), using 5% significance level.

Table 1 summarizes our sample construction. The original NAIC transaction data begin with 1,708,248 transactions on 119,790 bonds issued by 29,468 different issuers (based on the six-digit CUSIP). We exclude some transactions based on the criteria described in Table 1. The two biggest exclusions are for obvious errors (e.g., negative or missing prices or par values) and for non-secondary market transactions (e.g., redemptions and calls). The merge with FISD data results in a small additional exclusion of trades, mostly on bonds that are issued after August 2005 (end of our FISD data). Using the merged characteristics, we include only dollar-denominated corporate debentures and medium-term notes that have an issue size greater than \$50 million and do not have sinking funds provisions. The merge with SR insurance company data has little effect on the sample size since companies that are not covered by SR are small and trade relatively infrequently.

The final sample contains 483,710 transactions on 14,074 bonds issued by 4,228 issuers. There are 3,042 insurance companies behind the trades, of which 982 are life and health and 2,060 are property and casualty. From this sample, we then select 1,179 bonds that are downgraded from investment to speculative grades during the sample period. These bonds account for 66,021 transactions

by 2,451 insurance companies. We use this sample primarily for trading and volume analyses.

For studies of abnormal returns that require controls for issuer-specific information, we further merge each bond to the issuer's stock identifier, PERMNO, using CRSP's history name file. We obtain issuer stock identifiers for 1,043 of 1,179 downgraded bonds.¹³ Since we trace the abnormal returns over the period from 20 weeks before to 35 weeks after the downgrade, we only use the bonds that trade: (i) at least once before week –20, (ii) at least once between weeks –20 and 0, (iii) at least once between weeks 0 and +35, and (iv) at least once after week +35.¹⁴ In addition, we exclude the bonds that trade less than ten times during our sample period since these bonds are highly illiquid and may have unreliable prices. These screens leave us with 356 bonds issued by 173 issuers, accounting for 33,722 transactions.

Finally, to create an environment in which the information effects are minimal and the fire sales effects can

¹³ To ensure the reliability of our transaction prices, we compare them with those reported in TRACE for TRACE-eligible bonds. We find no systematic or significant differences.

¹⁴ We thank an anonymous referee for suggesting these additional screens to us.

Table 2

Bond and transaction descriptive information.

This table provides descriptive information regarding the samples used in the empirical analyses. The sample period is from 2001 to 2005. The total sample includes all transactions and underlying bonds that pass the screening process in Table 1 up to step (3). The “downgrade” sample includes only transactions on bonds that are downgraded from investment (BBB– and above) to non-investment (BB+ and below) grades during the sample period. Average issue size is calculated across bond issues, taking each issue as one observation. For each issue, insurance holding is the (quarterly time-series) median holding of all insurance companies together, measured as a percentage of the bond’s issue size. Average holding is then calculated across bond issues, taking each issue as one observation. Buy trades refer to transactions in which the reporting insurance companies buy (mostly from dealers). For transaction characteristics, the averages and medians are calculated across transactions, taking each transaction as one observation. The holding and transaction statistics for the “downgrade” sample are calculated separately for the periods before and after the downgrade.

	Always investment-grade	Downgrade		Always non-investment-grade	Not rated	Total
		Before downgrade	After downgrade			
Number of issues	8,081		1,179	4,634	180	14,074
Average issue size (\$ million)	381.15		416.88	277.17	303.71	348.92
Average insurance holding (% of issue size)	34.10	34.09	21.22	8.47	11.63	25.05
Number of trades	311,699	42,181	23,840	102,503	3,487	483,710
Number of trades per issue	39	36	20	22	19	34
% Buy trades	59.95	52.67	41.67	49.16	41.76	56.00
<i>Average trade characteristics</i>						
Price (per \$100 par)	105.38	99.16	90.32	96.01	80.49	101.93
Size (\$ million)	3.22	3.11	2.32	1.55	1.97	2.80
Age (years)	2.75	2.36	4.20	2.22	2.24	2.67
Maturity (years)	9.56	10.63	9.73	7.28	6.58	9.16
<i>Median trade characteristics</i>						
Price (per \$100 par)	104.10	100.45	96.25	100.96	94.23	102.70
Size (\$ million)	1.20	1.00	0.85	0.50	0.71	1.00
Age (years)	1.85	1.70	3.64	1.56	1.73	1.86
Maturity (years)	7.42	7.84	6.52	6.97	6.47	7.22

be most accurately identified, we follow the approach of Ambrose, Cai, and Helwege (2009) by studying the sample of bonds for which the bond issuers’ stock returns upon the downgrade are not significantly different from zero. Specifically, we compare the average daily stock return from day –1 to day +1 around the downgrade to the average historical stock return over 250 days (from days –280 to –31). Using 5% significance level, we identify 584 bonds as having no significant stock price reaction to the downgrade. Of these, 237 bonds trade in all four event periods discussed above. These bonds are issued by 117 unique issuers and account for 21,657 transactions.

2.2. Bond and transaction descriptive statistics

Table 2 describes important characteristics of the downgrade sample in comparison with bonds in other rating categories. Most bonds held and traded by insurance companies are investment-grade bonds. The average issue size of the downgraded bonds is close to that of the investment-grade issues and significantly larger than typical speculative-grade issues. This suggests that after the uncertainty surrounding the downgrade has been resolved, the downgraded bonds may be more liquid and frequently traded than typical non-investment-grade bonds. On average, insurance companies hold about 34% of investment-grade bonds (consistent with Schultz, 2001) and only 8% of non-investment-grade bonds. Clearly, insurance companies cut down their holdings of downgraded issues after the downgrade, from 34% to around 20%.

Table 2 also shows that investment-grade bonds trade more frequently and in larger size than non-investment-grade bonds. In total, the downgraded bonds trade even more often, likely due to activities around the downgrade events. Insurance companies tend to sell after the downgrade. This evidence, along with the large reduction in insurance holdings, suggests that bonds downgraded into non-investment ratings are likely to be heavily sold. Probably as a result of this selling, prices decline substantially after the downgrade.

Of the 1,179 downgraded bonds, 918 previously held a rating of BBB– and 209 a BBB. That is, almost 96% of the downgraded bonds were within two steps of the non-investment-grade and 78% were within one step. These statistics suggest that the downgrade should not have come as a big surprise to the market and thus its information content is likely to be small.¹⁵ Given that public information on the bond issuers is regularly updated, some investors may have anticipated a downgrade. Therefore, selling volume may pick up and prices may decline weeks before the official downgrade. This is consistent with the evidence in Table 2. Even before the downgrade, the soon-to-be downgraded bonds exhibit significantly lower ‘buy’ frequencies and lower transaction prices than other investment-grade bonds.

¹⁵ Many of these bonds had been put on the “watch list” months before the official downgrade. In addition, bond issuers report quarterly financial information that may allow investors to better anticipate the downgrade.

Table 3

Insurance company descriptive information.

This table provides descriptive information about the insurance companies that trade the bonds in the total sample and the “downgrade” sample. The sample period is from 2001 to 2005. The total sample includes all insurance companies behind the transactions that pass the screening process in Table 1 up to step (3). The “downgrade” sample includes only insurance companies behind the transactions on bonds that are downgraded from investment (BBB– and above) to non-investment (BB+ and below) grades during the sample period. The total number of trades only includes the trades that pass the sample screen in Table 1. All other variables are from the Street.com. The frequency is annual, and the variable values are for each year-end. The statistics reported are cross-sectional statistics based on the median value of each variable from 2000 to 2005. The detailed description of each variable is in Appendix A.

	Total sample								Downgrade sample			
	Life/health insurance (N=982)				Property/casualty insurance (N=2,060)				All insurance (N=2,451)			
	Mean	10th Pct	50th Pct	90th Pct	Mean	10th Pct	50th Pct	90th Pct	Mean	10th Pct	50th Pct	90th Pct
<i>Measures of size</i>												
Total number of bond trades (2001–2005)	293	5	73	795	95	3	34	218	195	10	60	473
Invested assets (\$ million)	2,536	7	128	4,818	448	7	48	755	1,367	9	94	2,018
Capital and surplus (\$ million)	270	3	31	589	187	4	24	303	251	4	35	463
<i>Measures of regulatory constraint</i>												
NAIC risk-based capital ratio (RBC ratio)	25.63	4.21	8.28	40.19	39.75	3.00	7.40	70.20	22.30	3.31	7.30	33.26
Risk-adjusted capital ratio 1 (RACR1)	2.28	0.86	1.93	3.92	8.29	0.59	1.80	9.80	4.21	0.67	1.80	4.90
<i>Measures of risk appetite</i>												
Holding of non-investment-grade bonds (% of invested assets)	2.66	0.00	1.63	6.65	0.72	0.00	0.00	2.26	1.60	0.00	0.40	5.10
Holding of risky assets (% of invested assets)	15.56	0.35	11.41	34.87	14.99	0.00	9.43	37.82	16.06	0.13	11.69	37.57

2.3. Insurance company descriptive statistics

Table 3 describes the financial position of insurance companies. The detailed definitions of different variables are provided in Appendix A. Life companies are larger both by total invested assets and by capital and surplus. The mean and median amount of capital and surplus are \$270 million and \$31 million for life companies and \$187 million and \$24 million for property companies. The difference in total invested assets is large, due to leverage, which is particularly high for life companies.

In subsequent cross-sectional analyses, we employ the following measures of regulatory constraint: the NAIC risk-based capital ratio (RBC ratio) and the SR's risk-adjusted capital ratios (RACR1). RBC ratio is the ratio of total adjusted capital to NAIC risk-based capital (RBC). RBC is the minimum capital that the insurance company must maintain given the inherent risks in its operations. RBC is calculated based on the NAIC's formula which reflects its assessment of risks of different asset classes and businesses. Insurance companies with higher RBC ratios are considered better capitalized and companies with RBC ratios below 2.0 are subject to supervisory interventions. RACR1 differs from RBC ratio in that the risk-adjusted capitals in the denominator of RACR1 are calculated based on SR's risk assessment. For example, bonds rated BB– to BB+ are subject to the capital requirement of 1.3% in the RBC formula for life companies but 2% in the RACR1 formula. The critical level for RACR1 is 1.0.

It is important to note that the insurance literature views RBC ratios as *indicative* of financial struggles rather than categorical (e.g., above or below 2.0 is not a sole criterion for regulatory scrutiny). Indeed, an insurer could

be in financial distress and still exceed its RBC requirement. Indeed, Klein and Wang (2009) demonstrate that only a small fraction of insurers fall below the “action” level RBC requirement, but that rating agency capital-adequacy tests are considerably more stringent.

In terms of regulatory constraints, life and property companies are similar at the median, but very different at the mean. The reason is that property companies in the right tail have extremely high capitalization ratios. Property companies hold significantly less non-investment-grade bonds, possibly because their claims are more uncertain and their business liabilities are relatively short-term compared to life companies. To the extent that the differences in distribution of both the constraint measures and the size measures are driven by the business nature, we need to treat the two types of insurance separately in studying how they make investment decisions.¹⁶ Due to the extreme positive skewness of size and capitalization ratios, we use logs and winsorize tails in our analysis.

Table 3 also reports statistics on investment risk-taking, which might reflect risk appetite or investment flexibility (e.g., due to company size or business risk). In general, life and property companies invest about the same percentage of their portfolios in “risky” assets. Consistent with insurance regulation, we classify the following assets as risky: non-investment-grade bonds, common and preferred stocks, non-performing mortgages, real estate, and other investments. The compositions of these assets, however, are different between the

¹⁶ However, life and property companies are similar with respect to the distribution of their liquidity index.

two types of insurance companies. Life companies invest more in non-investment-grade bonds than do property companies; the mean percentages are 2.66 and 0.72, respectively. This suggests that property companies may be selling the downgraded bonds more heavily around the downgrade.

Compared to an average insurance company in the total sample, an average company in the downgrade sample is slightly larger, has lower capitalization ratios, and holds slightly more non-investment-grade bonds and risky assets. Part of the reason is that the downgraded bonds are of lower quality even before the downgrade occurs and therefore are held more widely by life companies. That is, life companies account for a disproportionately larger portion of the downgrade sample than in the total sample. Again, the distinction between life and property companies is important in determining the ability and willingness to hold the downgraded bonds after the downgrade.

2.4. Control variables

We control for fundamental bond values in our analysis, using both market-wide information and bond-issuer specific information. The market-wide control variables include stock market returns, default-free bond yields, and corporate bond spreads. We use the CRSP value-weighted total return index as the market index. We use changes in the Fed's 10-year constant-maturity Treasury bond yield as proxy for the change in the default-free bond yield. We use the average yields of Barclay's U.S. high-yield and investment-grade corporate bond baskets as proxies for speculative-grade and investment-grade bond yields. These yields are then converted into spreads by subtracting the default-free rate. We use changes in these spreads as proxies for changes in corporate bond spreads. To adjust for differences in duration across bonds, we multiply changes in default-free yield and credit spreads by bond maturity to obtain approximate price returns (or percentage price changes) for each bond between two consecutive trade dates.

3. Regulatory constraints and fire sales

3.1. Probability of selling around the downgrade

Fire sales of downgraded corporate bonds should generally occur when such bonds are held by regulatory-constrained insurance companies. To identify fire sales and the associated price effects, we start at the individual insurance company level by investigating the effect of measured financial health on the propensity to sell the downgraded bonds. We model the probability that an insurance company (holding the bonds at the quarter-end preceding the downgrade) will sell the downgraded bonds during weeks 0 to +5 after the downgrade as a probit function:

$$\Pr(D_{i,j,k} = 1) = \Phi(\delta_0 + \delta_X X_{i,k} + \delta_Y Y_{i,j,k} + \delta_W W_k), \quad (1)$$

where $\Phi(\cdot)$ denotes the standard normal distribution, $D_{i,j,k}$ is a dummy variable that equals one if the insurance

company j (holding bond i) sells bond i during weeks +0 and +5 after event k and zero otherwise, $X_{i,k}$ is a vector of bond i 's static characteristics (e.g., issue size) and time-varying characteristics (e.g., bond age) at the time of event k , $Y_{i,j,k}$ is a vector of insurance company j 's characteristics before event k and possibly with regard to bond i (e.g., company j 's holding of bond i before event k), W_k is a vector of event-specific control variables (e.g., quarter-year dummies), and δ 's are the corresponding vectors of coefficients to be estimated.

We include proxies for regulatory constraints in $Y_{i,j,k}$ along with several control variables. To measure regulatory constraints, we first use the distinction between life companies and property companies. Property companies face relatively short-term and more uncertain liabilities than do life companies. The capital requirements for holding non-investment-grade bonds compared to other risky asset classes are also quite different, making it relatively less desirable for property companies to hold non-investment-grade bonds. Thus, we expect that property companies are more likely to immediately sell downgraded bonds. We also use the company's RBC ratio (both in logged linear form and as a step function) as a more direct measure of regulatory constraints. To avoid outliers, we estimate our probit models using only the insurance companies with RBC ratios (most recently before the event) between the 5th and 90th percentiles.

The first five columns of Table 4 report the coefficient estimates (Panel A) and the marginal effects (Panel B) for different probit specifications. The result in the first column is consistent with the fire sale hypothesis: property companies are significantly more likely to immediately sell downgraded bonds. The probability of selling is 2.6% higher for property companies compared to life companies. This effect is economically significant, given that the mean selling probability for the entire sample is only 7.1%.¹⁷

Next, we examine the effects of RBC ratio separately for life companies (columns Life (1) and Life (2)) and property companies (columns Property (1) and Property (2)) since they differ significantly in the nature of their business, liabilities, size, and importantly, the regulations that apply to them. We employ 16,007 observations for the life sample and 6,263 observations for the property sample. For both types of companies, we find that *more constrained* insurers are more likely to immediately sell the downgraded bonds. The effect of regulatory constraint on selling probability is both economically and statistically significant. Considering the property sample as an example, if we take a company and increase its RBC ratio (holding everything else constant) from the 25th to the 75th percentiles of the sample (hence, making it less regulatory constrained), this company's probability of selling will decrease by 1.8%. This effect is sizeable considering that the mean selling probability for property companies is 9.5%.¹⁸

¹⁷ We also estimate the probit model using RACR1 as an alternative measure of regulatory constraints. The results, unreported for brevity, are largely in line with those reported in Table 4.

¹⁸ Our sample includes insurance companies with RBC ratios both above and below the regulatory action level of 2.0. Some financially impaired companies are also included.

Table 4

Probability of selling around the downgrade.

This table reports probit estimates for the effects of regulatory constraints on the probability that an insurance company will sell the downgraded bonds during (i) event weeks 0 to +5 and (ii) event weeks –10 to 0. The sample period is from 2001 to 2005. The downgrade is from investment to non-investment grades. The dependent variable is a dummy that equals one if the insurance company (holding the bond) sells the downgraded bond during the period of interest, and zero otherwise. Panel A reports the coefficient estimates. Panel B reports the corresponding marginal effects evaluated at the sample means of the explanatory variables. In columns Life vs. property (1) and Life vs. property (2), the difference in selling propensity between life and property companies is analyzed. In columns Life (1) to Life (3), only life companies are used in the estimation. In columns Property (1) to Property (3), only property companies are used in the estimation. All models include state and year-quarter dummies. White's robust standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively. Variable definitions are in Appendices A and B.

	Selling during weeks 0 to +5					Selling during weeks –10 to 0		
	Life vs. property (1)	Life (1)	Life (2)	Property (1)	Property (2)	Life vs. property (2)	Life (3)	Property (3)
Panel A: Coefficient estimates								
<i>Regulatory constraints</i>								
Property insurance dummy	0.183*** (0.028)					0.232*** (0.026)		
ln(RBC ratio)		–0.262*** (0.060)		–0.202*** (0.070)			–0.129** (0.056)	–0.115* (0.062)
Dummy: RBC ratio ≤ 25th pct			0.146*** (0.048)		0.204*** (0.062)			
Dummy: 25th pct < RBC ratio ≤ 50th pct			0.184*** (0.044)		0.185** (0.072)			
Dummy: 50th pct < RBC ratio ≤ 75th pct			0.091** (0.045)		0.119* (0.071)			
<i>Insurance company controls</i>								
ln(Capital & surplus)		–0.009 (0.009)	–0.012 (0.009)	0.007 (0.014)	0.007 (0.014)		–0.009 (0.009)	0.014 (0.013)
Risky asset holding (% of portfolio)		0.001 (0.002)	0.001 (0.002)	–0.007*** (0.002)	–0.007*** (0.002)		–0.002 (0.002)	–0.005*** (0.001)
<i>Bond controls</i>								
Lower than BB+ dummy	0.090*** (0.026)	0.070** (0.032)	0.070** (0.032)	0.178*** (0.048)	0.178*** (0.048)	0.138*** (0.024)	0.152*** (0.031)	0.169*** (0.043)
ln(Age in years)	–0.120*** (0.014)	–0.123*** (0.018)	–0.124*** (0.018)	–0.111*** (0.026)	–0.110*** (0.026)	–0.108*** (0.013)	–0.132*** (0.017)	–0.061** (0.024)
ln(Issue size)	0.081*** (0.016)	0.076*** (0.020)	0.075*** (0.020)	0.076*** (0.029)	0.076*** (0.029)	0.152*** (0.015)	0.139*** (0.019)	0.150*** (0.025)
Observations	23,690	16,007	16,007	6,263	6,263	24,180	15,906	6,540
Pseudo R-squared	0.054	0.069	0.069	0.069	0.070	0.082	0.098	0.080
Log likelihood	–6,373	–4,081	–4,081	–2,043	–2,042	–7,584	–4,726	–2,546
Panel B: Marginal effects								
<i>Regulatory constraints</i>								
Property insurance dummy	0.026*** (0.004)					0.040*** (0.005)		
ln(RBC ratio)		–0.031*** (0.007)		–0.034*** (0.012)			–0.019** (0.008)	–0.024* (0.013)
Dummy: RBC ratio ≤ 25th pct			0.019*** (0.007)		0.035*** (0.011)			
Dummy: 25th pct < RBC ratio ≤ 50th pct			0.023*** (0.006)		0.034** (0.014)			
Dummy: 50th pct < RBC ratio ≤ 75th pct			0.011* (0.006)		0.021 (0.013)			
<i>Insurance company controls</i>								
ln(Capital & surplus)		–0.001 (0.001)	–0.001 (0.001)	0.001 (0.002)	0.001 (0.002)		–0.001 (0.001)	0.003 (0.003)
Risky asset holding (% of portfolio)		0.000 (0.000)	0.000 (0.000)	–0.001*** (0.000)	–0.001*** (0.000)		–0.000 (0.000)	–0.001*** (0.000)
<i>Bond controls</i>								
Lower than BB+ dummy	0.012*** (0.004)	0.008** (0.004)	0.008** (0.004)	0.031*** (0.008)	0.030*** (0.008)	0.023*** (0.004)	0.022*** (0.005)	0.036*** (0.009)
ln(Age in years)	–0.016*** (0.002)	–0.015*** (0.002)	–0.015*** (0.002)	–0.019*** (0.004)	–0.019*** (0.004)	–0.017*** (0.002)	–0.019*** (0.002)	–0.013** (0.005)
ln(Issue size)	0.011*** (0.002)	0.009*** (0.002)	0.009*** (0.002)	0.013*** (0.005)	0.013*** (0.005)	0.024*** (0.002)	0.020*** (0.003)	0.032*** (0.005)

We also consider possible non-linearities in the relationship between RBC ratio and probabilities of selling. We introduce a series of dummy variables: for an RBC ratio below the 25th percentile (RBC ratio of 5.0), an RBC ratio between the 25th and 50th percentiles (RBC ratio of 6.3 at the 50th percentile), and an RBC ratio between the 50th and 75th percentiles (RBC ratio of 8.0 at the 75th percentile).¹⁹ Consistent with the results from linear specifications above, insurance companies with RBC ratios below the median are significantly more likely (2% more for life insurers and 3.5% more for property insurers) to sell the downgraded bonds than those with RBC ratios in the top quartile.

The impact of regulatory constraints on the probability of immediate selling is robust to the inclusion of a host of control variables that may be associated with selling for unrelated reasons. First, all specifications include year-quarter dummies and U.S. state (of incorporation) dummies. Second, significant variables include the proportion of risky assets in the portfolio, a dummy variable that indicates whether the downgrade is into a rating lower than BB+, and the bond's age and issue size. These reported coefficients on the control variables are as expected. Among these effects, insurance companies are more likely to sell younger bonds and bonds with larger issue size. Edwards, Harris, and Piwowar (2007), among others, find that bid-ask spreads increase with bond age and decrease with bond issue size. One interpretation of our results is that insurance companies actively try to minimize price impact by avoiding the sale of illiquid bonds. Thus, the price effects of downgrades may actually be *larger* for relatively more liquid bonds, as they are more heavily sold.

In light of the evidence that bond prices (Hite and Warga, 1997; Steiner and Heinke, 2001; among others) and credit default swaps prices (Hull, Predescu, and White, 2004; Norden and Weber, 2004) anticipate bond downgrades, we assume that rating downgrades may be somewhat predictable and hypothesize that constrained insurers, trying to avoid selling at fire sale prices, may start selling weeks before the actual downgrade. To examine this possibility, we also estimate our probit specification for selling over the period from weeks –10 to 0. Consistent with our earlier results, the last three columns of Table 4 show that more constrained insurers are more likely to sell in the period leading up to the downgrade. This finding holds when comparing property and life insurers, and separately for both life and property companies. In sum, regulatory constraints are an important determinant of individual insurance companies' selling of downgraded bonds around the downgrade announcement. Whether the forced selling of regulatory-constrained insurers we document generates significant price effects will depend on whether the selling by these insurers is representative of the insurance industry's aggregate trading activity. We turn to this subject in the next section.

3.2. Trading volume around the downgrade

In Table 5, we investigate the total volume (sales plus buys) and net volume (sales less buys) of downgraded bonds (1) by all insurance companies, and separately (2) by insurance companies with RBC ratios below the sample median. Based on our probit analysis, we consider the latter regulatory-constrained. If the fire sales hypothesis is correct, then constrained insurers should account for a disproportionately large fraction of selling activity around the rating downgrade.

Panel A shows that before the downgrade, the total volume is approximately \$0.9–1 million per (downgraded) bond per week, and more than 73% of this volume is generated by constrained insurance companies.²⁰ In week 0 (weeks from +1 to +5), an average of more than \$2.4 million per bond (\$1 million per bond per week) is transacted and almost 74% (75%) comes from constrained insurers. The dollar volume decreases after week +5 but the importance of the constrained insurers remains elevated. A similar picture emerges when we consider net volume, only that the importance of constrained insurers gets larger in such an analysis. For example, in the downgrade week, the average net volume is about –\$1.3 million per bond (net sales); constrained insurers account for more than \$1 million of the net sales. Fig. 1 plots the average net volume per bond by event week around the downgrade. Clearly, starting from around week –7, the net selling volume of relatively constrained insurers is several times higher than that of unconstrained insurers. This difference becomes increasingly apparent in the period after the downgrade. Forced selling of constrained insurers appears to be representative of the market starting from a few weeks before to several weeks after the downgrade.

The last column of Table 5 Panel A tests whether in each event period the number of downgraded bonds that are net sold by the insurance industry is significantly higher than those that are net bought. Clearly, the insurance industry turns from net buyers to net sellers of the downgraded bonds after event week –20.

In Panel B of Table 5, we compare the net dollar volume (from weeks –10 to +10) of bonds with the average selling probability above the sample median (first three columns) and those with the average selling probability below the sample median (last three columns). We calculate the bond-level average selling probability as follows. For each insurer holding a downgraded bond, we calculate the predicted immediate (weeks 0 to +5) selling probability using our estimated probit specifications labeled Life (1) and Property (1). We then aggregate the insurer-level selling probability into the bond-level selling probability by averaging across all insurers that hold the bond at the end of the quarter preceding the downgrade. Therefore, this bond-level average selling probability proxies for regulatory-induced selling pressure. We view the bonds with average selling probability above the median as the high regulatory-constraint group

¹⁹ These percentiles of RBC ratio are based on the sample of insurance companies holding the downgraded bonds at the end of quarter preceding the downgrade, with the lowest 5% and highest 10% of RBC ratios dropped.

²⁰ Ambrose, Cai, and Helwege (2008) investigate the overall insurance industry's selling volume of the bonds that are downgraded from investment to non-investment grades.

Table 5

Insurance companies' trading volume around the downgrade.

This table reports insurance companies' trading volume of the downgraded bonds. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. A total of 1,179 downgraded bonds (sample 4 in Table 1) are considered. Week 0 is the downgrade announcement week. For each bond in each event week, the total volume is calculated as the sum of the buy volume and the sell volume, while the net volume is calculated as the buy volume minus the sell volume. In columns "All insurance," all insurance companies' trades are included while in columns "Below median RBC ratio," the volumes reflect only trades by insurance companies with RBC ratio below the sample median. Panel A reports the average total and net volumes across all bond-weeks during the defined period. The last column reports the *p*-values from one-sided sign tests of the null hypothesis that the number of bonds that are net sold is less than or equal to the number of bonds that are net bought. Panel B compares the average net volume between two groups of bonds with the average selling probability above and below the sample median. For each insurance company, the selling probability is calculated based on the models Life (1) and Property (1) in Table 4. The bond-level average selling probability is calculated by averaging the selling probability across all insurance companies holding the bond at the end of the quarter preceding the downgrade.

5-Week period	Average total volume (\$ million/bond/week)		Below median RBC ratio as % of total	Net volume (\$ million/bond/week)		Net sell frequency $\leq$ net buy frequency? Sign test (<i>p-value</i>)
	All insurance	Below median RBC ratio		All insurance	Below median RBC ratio	
<i>Panel A: Volume by event period</i>						
[−30, −26]	1.019	0.830	72.5	0.020	−0.008	(0.991)
[−25, −21]	0.901	0.730	75.9	0.043	0.025	(0.835)
[−20, −16]	0.925	0.736	76.8	−0.061	−0.041	(0.002)
[−15, −11]	0.951	0.784	75.4	−0.180	−0.149	(0.000)
[−10, −6]	0.959	0.793	75.7	−0.307	−0.225	(0.000)
[−5, −1]	0.993	0.772	73.1	−0.433	−0.299	(0.000)
Week 0	2.482	1.976	73.8	−1.344	−1.058	(0.000)
[1, 5]	1.062	0.834	74.4	−0.492	−0.408	(0.000)
[6, 10]	0.883	0.732	78.2	−0.460	−0.382	(0.000)
[11, 15]	0.646	0.542	79.1	−0.289	−0.248	(0.000)
[16, 20]	0.597	0.475	76.9	−0.229	−0.199	(0.000)
[21, 25]	0.549	0.404	74.1	−0.315	−0.228	(0.000)
[26, 30]	0.509	0.420	79.4	−0.236	−0.213	(0.000)
[31, 35]	0.499	0.369	75.4	−0.186	−0.143	(0.000)
5-Week period	<i>Above-median average selling probability</i>			<i>Below-median average selling probability</i>		
	All insurance	Below median RBC ratio	Below median RBC ratio as % of total	All insurance	Below median RBC ratio	Below median RBC ratio as % of total
<i>Panel B: Net volume for bonds with average selling probability above and below median</i>						
[−10, −6]	−0.577	−0.401	73.5	−0.189	−0.148	77.8
[−5, −1]	−1.184	−0.875	71.5	−0.110	−0.052	75.0
Week 0	−3.201	−2.477	73.3	−0.553	−0.453	74.1
[1, 5]	−1.044	−0.814	72.7	−0.258	−0.235	75.9
[6, 10]	−0.896	−0.715	77.4	−0.274	−0.240	78.8

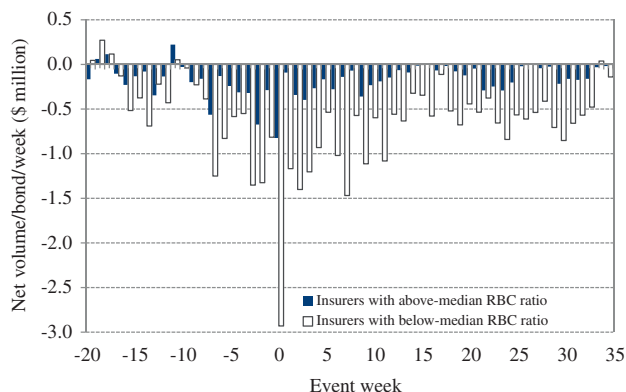


Fig. 1. Net volume by RBC ratio. This figure plots insurance companies' net trading volume of bonds that are downgraded from investment to non-investment grade. The sample period is from 2001 to 2005. A total of 1,179 downgraded bonds (sample 4 in Table 1) are considered. Insurance companies are split into two groups at the median of their RBC ratios. Week 0 is the downgrade announcement week. For each bond in each event week, the net volume is calculated as the buy volume minus the sell volume. No trading counts as zero net volume. The net volume is calculated separately for each group of insurance companies. The average net volume is obtained by averaging the net volume across all downgraded bonds.

and those with average selling probability below the median as the low regulatory-constraint group. Consistent with the fire sales hypothesis, the net selling volume by all insurers (the first and fourth columns) is significantly larger for the high-constraint bonds. However, even for the low-constraint bonds, a disproportionately large share of the net volume comes from relatively constrained insurers. These results suggest that forced selling may represent the selling activity in both the high- and low-constraint groups of bonds, but the high-constraint group may suffer larger price effects due to the significantly larger overall sale volume.

4. Price effects of fire sales

4.1. Median cumulative abnormal returns

We next proceed to investigate whether the selling pressure shown in Section 3 produces significant price effects. We study cumulative abnormal returns from 20 weeks before to 35 weeks after the downgrade. To disentangle price pressure from information revelation, we follow the approach used by Mitchell, Pulvino, and Stafford (2004) and Coval and Stafford (2007). Specifically, we look for evidence of price declines followed by significant reversals. If insurance companies' trading around the downgrade is motivated by information, then transaction prices (and fundamental values) should drop during the period of heavy sales and stabilize permanently at the lower level. On the other hand, if insurance companies have to sell immediately due to regulatory constraints, then these companies may face fire sale prices. In this case, a drop in transaction prices should be followed by a series of positive abnormal returns compensating liquidity providers for (eventually) stepping in.

We measure abnormal returns using a simple market model that controls for both fundamental value changes and transaction costs. Our model is similar in spirit to the models used by Bessembinder, Maxwell, and Venkataraman (2006) and Schultz (2001), among others. We start by assuming that the observed trade price of bond i on date t , denoted by $P_{i,t}$, equals the fundamental bond value $E_t(V_i)$ adjusted up or down by the percentage half spread $A_{i,t}$:

$$P_{i,t} = (1 + A_{i,t} Q_{i,t}) \times E_t(V_i), \quad (2)$$

where $Q_{i,t}$ denotes an indicator variable that equals one (-1) if the trade on date t of bond i is an insurance company's buy (sell). The logged price return (or change in logged price) from one trade date to the next is thus given by

$$\ln(P_{i,t}/P_{i,t-1}) \cong (A_{i,t} Q_{i,t} - A_{i,t-1} Q_{i,t-1}) + \ln(E_t(V_i)/E_{t-1}(V_i)). \quad (3)$$

We specify the half spread as a function of trade size $S_{i,t}$ in the form²¹

$$A_{i,t} = \gamma_0 + \gamma_1 \ln(S_{i,t}) + \eta_{i,t}, \quad (4)$$

²¹ Virtually all studies of corporate bond trading costs (see Bessembinder, Maxwell, and Venkataraman, 2006; Edwards, Harris, and Piwowar, 2007; Goldstein, Hotchkiss, and Sirri, 2007) find that trading costs can be extremely large for small trades and that trading costs decline sharply with trade size. Bao and Pan (2008) find that

where $E(\eta_{i,t}) = 0$. Guided by the structural model of Merton (1974), we specify the change in fundamental value as a function of (i) changes in the government bond yield, (ii) changes in the investment-grade corporate spread, (iii) changes in the high-yield corporate spread, (iv) aggregate stock market returns, and (v) bond-issuer stock returns following the spirit of Ambrose, Cai, and Helwege (2009). We account for the duration effect by making the government and corporate bond spread changes proportional to the remaining years to maturity of bond i . Let these variables from trade dates $t-1$ to t be denoted by $Z_{i,t-1,t}$. Assuming a linear relationship, we then have

$$\ln(E_t(V_i)/E_{t-1}(V_i)) = \alpha + \beta Z_{i,t-1,t} + v_{i,t-1,t}, \quad (5)$$

where $E(v_{i,t-1,t}) = 0$. Substituting (3) and (4) into (2), we obtain

$$\ln(P_{i,t}/P_{i,t-1}) \cong \alpha + \beta Z_{i,t-1,t} + \gamma_0(Q_{i,t} - Q_{i,t-1}) + \gamma_1(Q_{i,t} \ln(S_{i,t}) - Q_{i,t-1} \ln(S_{i,t-1})) + \varepsilon_{i,t-1,t}. \quad (6)$$

By construction, $\varepsilon_{i,t-1,t} = v_{i,t-1,t} + Q_{i,t} \eta_{i,t} - Q_{i,t-1} \eta_{i,t-1}$ and $E(\varepsilon_{i,t-1,t}) = 0$. We further make the usual assumption that the right-hand side-variables and ε are uncorrelated.

We define the event as the first downgrade announcement and the event period (not used for parameter estimation) as the period from event weeks -20 to $+35$. We estimate Eq. (6) using the data for the period before week -20 to obtain the abnormal returns from weeks -20 to -1 .²² We do so separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and pre-downgrade rating at BBB-, (ii) maturity at downgrade > 5 years and pre-downgrade rating at BBB-, (iii) maturity at downgrade ≤ 5 years and pre-downgrade rating above BBB-, and (iv) maturity at downgrade > 5 years and pre-downgrade rating above BBB-, essentially imposing the restriction that the parameters are the same for all bonds in the same group. Our grouping is motivated by the balance between empirical power and theoretical accuracy. Since bonds do not trade often, parameter estimates obtained separately for each bond are potentially noisy and unreliable.²³ Essentially, we need a sufficient number of transactions in each bond group to obtain reliable estimates. On the other hand, the structural model of Merton (1974) predicts a non-linear relationship between bond and stock prices; so, assuming that bonds of different ratings (hence different distance to default) load equally on aggregate stock market and/or

(footnote continued)

corporate bond prices exhibit excess volatility compared to what is predicted by Merton's (1974) model and that the degree of excess volatility is negatively related to measures of bond liquidity including the average trade size. In extracting abnormal returns, controlling for the effect of trade size on trading costs is equivalent to assigning some potentially large residuals associated with smaller trades to the non-event-related time-invariant microstructure effects.

²² We choose the preceding estimation period to end quite early in order to avoid the contamination of abnormal price movements around the downgrade. Hite and Warga (1997) and Steiner and Heinke (2001), among others, show that prices anticipate bond downgrades.

²³ For example, bonds that are downgraded from an investment-grade rating to a non-investment-grade rating only trade 27 times at the median and 56 times, on average, in our sample period. Edwards, Harris, and Piwowar (2007) estimate a similar model separately for each bond but they use TRACE data which are much denser.

issuer-stock returns would be inaccurate. We also use bond maturity to adjust for duration effects as we convert interest rate and spread changes to price returns. The difference between modified duration and maturity grows as maturity increases, and so treating maturity adjustments the same way for all bonds would be incorrect too. For these reasons, we settle for this broad grouping based on credit rating and maturity.

Intuitively, the relationship (6) changes around the event in the sense that bonds in the lower rating categories are relatively more risky and equity-like, and therefore load more on the change in the high-yield spread and the stock market return. Moreover, [Edwards, Harris, and Piwowar \(2007\)](#) find that transaction costs are substantially larger for high-yield bonds than for investment-grade bonds. Therefore, we re-estimate (6) using the data starting in week +36 to obtain the abnormal returns from weeks 0 to +35. Again, for the reasons discussed above, we estimate the parameters separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and post-downgrade rating at BB+, (ii) maturity at downgrade > 5 years and post-downgrade rating at BB+, (iii) maturity at downgrade ≤ 5 years and post-downgrade rating below BB+, and (iv) maturity at downgrade > 5 years and post-downgrade rating below BB+.

The abnormal return $AR_{i,t-1,t}$ is the error term $\varepsilon_{i,t-1,t}$ for observations in the event period calculated using the parameter estimates from the estimation period. Since we have controlled for both market-wide and issuer-specific information, AR should reflect only the liquidity (temporary) component associated with price concessions. To obtain the cumulative abnormal return $CAR_{i,t}$, we accumulate $AR_{i,t-1,t}$ up to date t . We use the median abnormal return in the case that multiple trades in the same bond take place on the same day at different prices. To reduce noise, we further aggregate CAR for each bond by week relative to the downgrade event. We normalize the CAR for each bond to zero at week -20. Finally, we calculate the median cumulative abnormal return, MCAR, as the median of the normalized CARs across all bonds in a particular group that trade in each event week.²⁴ We only use trade prices in calculating MCARs; bonds that do not trade in a particular event week do not enter the median calculation for that week.

Not all bonds enter the MCAR calculation. Given our return specification (6), only bonds that trade on at least two days before week -20 (after week +35) enter the estimation of parameters for the pre-downgrade (post-downgrade) period. In addition, we exclude the bonds that trade less than ten times during our sample period since these bonds are highly illiquid and may have unreliable prices. Finally, we only use the bonds that trade: (i) at least once before week -20, (ii) at least once between weeks -20 and 0, (iii) at least once between

weeks 0 and +35, and (iv) at least once after week +35. The main reason is that we want to more precisely estimate the entire price trajectory around the downgrade. These screens leave us with 356 bonds issued by 173 issuers, accounting for 33,722 transactions (Sample 5.1 in [Table 1](#)).

To see whether the selling pressure documented earlier translates into distinguishable price effects, we compare MCARs for bonds with average selling probability above the median (high regulatory constraint) with those for bonds with average selling probability below the median (low regulatory constraint). [Fig. 2](#) plots the MCARs for the two constraint groups by event week. [Table 6](#) reports the MCARs by five-week period and tests whether the MCARs for the two constraint groups are different.

Consistent with the fire sales hypothesis, [Fig. 2](#) shows that the bonds with high regulatory-induced selling probability experience larger deviations from fundamental values, as exhibited by larger price declines and subsequent reversals. As [Table 6](#) reports, during the first five weeks after the downgrade, the MCARs are significantly more negative for the high selling probability group than for the low selling probability group (-8.7% vs. -2.8%). This difference gets progressively smaller after week +5 and largely disappears after week +30 when prices appear to fully recover and stabilize.²⁵ Thus, the most plausible interpretation is that fire sales caused by regulatory constraints lead to price declines beyond fundamental values, equivalent in magnitude to the difference in the MCARs between the high and low regulatory-induced selling probability groups (around 6% and 7% from week 0 to week +5).

[Table 6](#) contains other important results that also corroborate the fire sales story. First, the difference in the MCARs between the high regulatory-constraint bonds and the low regulatory-constraint bonds turn negative over the period from around week -10 to week -6. This evidence is largely consistent with the probit results in [Table 4](#) where we show that regulatory constraints affect an insurer's selling probability even before the actual downgrades. Second, the documented deviations from fundamental values are temporary, completely disappearing by weeks +30 to +35. This suggests that the price effects are largely due to insufficient liquidity (possibly due to lack of sufficient buyers). As more buyers arrive, transaction prices slowly revert to the fundamental values. Together, these results are consistent with the fire sales hypothesis and the associated role for regulation.

²⁴ Since two corporate bond trades may be weeks apart, we do not have abnormal returns for every bond in each event week. We cannot calculate the weekly median or average first and then accumulate the aggregate statistic over time (as in [Coval and Stafford, 2007](#)). Also, the number of bonds that are used in the calculation of MCARs varies from week to week.

²⁵ The same qualitative patterns, albeit larger in magnitude, are observed for the average cumulative abnormal returns (ACARs), which are not reported for brevity. Due to the large outliers on the negative side, we consider MCARs more stable and appropriate, and use them only from this point onwards. [Bessembinder, Kahle, Maxwell, and Xu \(2009\)](#) find that skewness in abnormal returns leads to excessive Type I errors (over-rejections of the null) in the t -statistics (mean tests). They also find that the mean of abnormal returns tends to be noisy and to vary with specifications, while the median is more robust. However, since the mean better reflects the experience of traders in aggregate, they suggest that one should evaluate test statistics based on both mean and median returns.

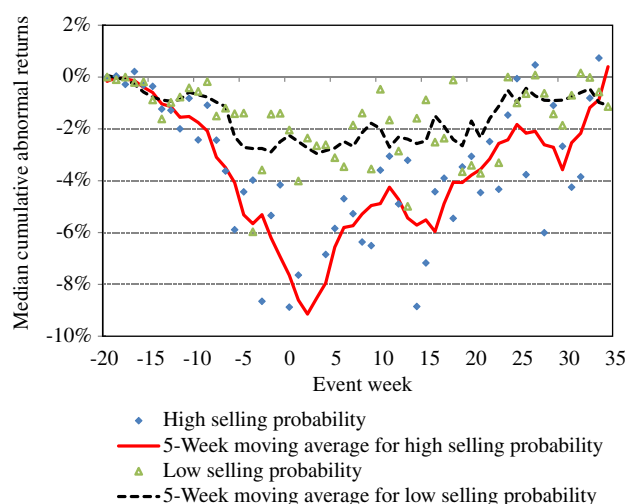


Fig. 2. Median cumulative abnormal returns by average selling probability. This figure plots the median cumulative abnormal returns (MCARs) for downgraded bonds grouped by bond-level average selling probability. Event week is in the X-axis and week 0 is the downgrade announcement week. For each insurance company, the selling probability is calculated based on the models Life (1) and Property (1) in Table 4. The bond-level average selling probability is calculated by averaging the selling probability across all insurance companies holding the bond at the end of the quarter preceding the downgrade. MCARs are calculated by the model-adjustment methodology. The model adjusts for both fundamental price movement and transaction costs. Included variables are duration-adjusted return on 10-year Treasury note, duration-adjusted return on Barclays investment-grade corporate bond spread, duration-adjusted return on Barclays high-yield corporate bond spread, CRSP value-weighted return, bond issuer's stock return, current and lagged buy/sell indicator variables, and the interaction between buy/sell indicator variables and logged trade size. The period before week -20 is used as the estimation period to obtain abnormal returns from weeks -20 to -1 . The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and pre-downgrade rating at BBB $-$, (ii) maturity at downgrade > 5 years and pre-downgrade rating at BBB $-$, (iii) maturity at downgrade ≤ 5 years and pre-downgrade rating above BBB $-$, and (iv) maturity at downgrade > 5 years and pre-downgrade rating above BBB $-$. The period starting in week $+36$ is used as the estimation period to obtain abnormal returns from weeks 0 to $+35$. The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and post-downgrade rating at BB $+$, (ii) maturity at downgrade > 5 years and post-downgrade rating at BB $+$, (iii) maturity at downgrade ≤ 5 years and post-downgrade rating below BB $+$, and (iv) maturity at downgrade > 5 years and post-downgrade rating below BB $+$. For each bond, the abnormal returns are accumulated over time, and normalized to zero in week -20 . MCARs are measured as across-bond across-transaction medians of cumulative abnormal returns for each week. The calculation uses only actual trade prices on bonds that trade at least once before week -20 , at least once between weeks -20 and 0, at least once between weeks 0 and $+35$, and at least once after week $+35$, i.e., Sample 5.1 in Table 1.

4.2. Insurance industry distress and potential buyers' capital

As Shleifer and Vishny (1992) suggest, fire sales are likely to happen when both sellers and buyers are constrained. In this environment, many constrained sellers are competing to sell to just a handful of healthy buyers. The buyers have all the bargaining power, forcing the sellers to sell at prices significantly below fundamental values. In this section, we attempt to identify the environment in which fire sales are most likely.

We measure the constraints on the sell side by the degree of distress of the insurance industry as a whole. We use the "Impairment Ratio,"²⁶ which measures the number of insurance companies becoming "impaired" in a given year as a ratio (percentage) of the total number of insurance companies. Insurance companies are considered impaired when they have had an official regulatory action taken against them by regulators in their states of domicile. State actions limit the insurers' freedom to conduct business normally, and are usually taken when the insurers are (or are close to being) insolvent. Columns 1–3 of Table 7 show the annual impairment percentages

for the life and health sector, the property and casualty sector, and the insurance industry as a whole in terms of asset-weighted average. Clearly, both sectors are comparatively under stress (high impairment percentages) in 2001–2002. The life and health sector is also under stress in 2005.

In Table 7, we also relate the net trading volume of downgraded bonds to the annual impairment percentages. For each bond, the net volume is calculated as buy volume minus sell volume during the period from weeks -5 to $+5$, divided by the bond issue size. In columns 4–6, depending on specification, we regress the net volume on downgrade year dummies, bond-level average selling probability, and bond characteristics (logged age, logged issue size, and logged maturity). For the bonds downgraded during the periods of high impairment (2001–2002 and 2005), insurance companies sell about 2–3% of the issue size from weeks -5 to $+5$. The net volume is about zero for the bonds downgraded in 2003–2004 when the insurance industry is not under stress. These results are robust in all specifications. Fig. 3 shows visually by downgrade quarter, the positive relationship between the impairment percentage and insurance companies' selling volume.

Interestingly, column 6 of Table 7 also shows that the time variation in observed selling activity is not the same

²⁶ The impairment ratio data were provided by the Insurance Industry Institute (www.iii.com).

Table 6

Cumulative abnormal returns of downgraded bonds by average selling probability.

This table reports the median cumulative abnormal returns (MCARs) for the downgraded bonds grouped by the average probability of selling during weeks 0 and +5 after the downgrade. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. A total of 1,179 downgraded bonds (sample 4 in Table 1) are considered. Week 0 is the downgrade announcement week. For each insurance company, the selling probability is calculated based on the models Life (1) and Property (1) in Table 4. The bond-level average selling probability is calculated by averaging the selling probability across all insurance companies holding the bond at the end of the quarter preceding the downgrade. The abnormal return for a bond from $t-1$ to t is measured as the residual ε_t from the model:

$$\ln(P_{i,t}/P_{i,t-1}) \cong \alpha + \beta Z_{i,t-1,t} + \gamma_0(Q_{i,t} - Q_{i,t-1}) + \gamma_1(Q_{i,t} \ln(S_{i,t}) - Q_{i,t-1} \ln(S_{i,t-1})) + \varepsilon_{i,t-1,t},$$

where Z is a vector of information variables that impact the bond's fundamental value, including duration-adjusted return on 10-year Treasury note, duration-adjusted return on Barclays investment-grade corporate bond spread, duration-adjusted return on Barclays high-yield corporate bond spread, CRSP value-weighted return, and bond issuer's stock return. Q is an indicator variable equal to one if the transaction is a buy and -1 if it is a sell. S denotes trade size (par value). The parts associated with Q are included to control for the normal price impact of a trade. The model is estimated using the data for the period before week -20 to obtain the abnormal return from weeks -20 to -1 . The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and pre-downgrade rating at BBB-, (ii) maturity at downgrade > 5 years and pre-downgrade rating at BBB-, (iii) maturity at downgrade ≤ 5 years and pre-downgrade rating above BBB-, and (iv) maturity at downgrade > 5 years and pre-downgrade rating above BBB-. The model is then re-estimated using the data starting in week $+36$ to obtain the abnormal return from weeks 0 to $+35$. The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and post-downgrade rating at BB+, (ii) maturity at downgrade > 5 years and post-downgrade rating at BB+, (iii) maturity at downgrade ≤ 5 years and post-downgrade rating below BB+, and (iv) maturity at downgrade > 5 years and post-downgrade rating below BB+. The abnormal returns are then accumulated over time. The cumulative abnormal return for each bond is then normalized to zero in week -20 . MCARs are measured as across-bond across-transaction medians of cumulative abnormal returns for each five-week period relative to the event. Only 356 downgraded bonds in Sample 5.1 in Table 1 are used. The t -statistics are for the tests of (i) null hypothesis that MCAR equals zero and (ii) null hypothesis that MCARs for the two average selling probability groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds with average selling probability above the median).

5-Week period	Above-median average selling probability		Below-median average selling probability		Difference	
	(1) MCAR	H0: MCAR=0 t -statistic	(2) MCAR	H0: MCAR=0 t -statistic	(1)-(2) MCAR	H0: (1)=(2) t -statistic
[-20, -16]	0.000	(0.00)	0.000	(0.00)	0.000	(0.00)
[-15, -11]	-0.814	(-4.07)	-0.879	(-4.25)	0.065	(0.16)
[-10, -6]	-2.987	(-6.18)	-0.967	(-3.38)	-2.020	(-3.31)
[-5, -1]	-5.323	(-6.77)	-2.147	(-3.65)	-3.176	(-3.11)
Week 0	-8.877	(-5.00)	-2.045	(-2.62)	-6.832	(-3.87)
[1, 5]	-8.680	(-9.22)	-2.824	(-4.49)	-5.857	(-5.27)
[6, 10]	-5.344	(-9.01)	-2.334	(-3.32)	-3.009	(-3.37)
[11, 15]	-4.276	(-5.35)	-2.430	(-3.83)	-1.846	(-1.72)
[16, 20]	-3.618	(-5.12)	-2.498	(-4.68)	-1.120	(-1.61)
[21, 25]	-2.233	(-3.42)	-2.430	(-3.23)	0.198	(0.22)
[26, 30]	-3.880	(-4.73)	-1.063	(-1.48)	-2.817	(-2.64)
[31, 35]	-1.101	(-0.75)	-0.527	(-0.99)	-0.574	(-0.14)

as the cross-sectional variation driven by the financial health of individual companies holding each bond. Both the insurance industry's overall stress and the bond-level average selling probability seem to be important in explaining the selling activity for each downgraded bond. In sum, we can reasonably identify 2001, 2002, and 2005 as periods in which insurance companies are constrained and heavily sell downgraded issues.

As a measure of constraints on the buy side, we use the total assets of high-yield bond mutual funds and high-yield and distress-focused hedge funds. Our assumption is that these are potential buyers of downgraded bonds and the more assets they have, the more flexible they are to step in and buy the downgraded bonds from insurance companies. We collect monthly data on assets under management (AUM) of high-yield bond mutual funds from the CRSP Mutual Fund Database (using Lipper codes to identify fund objective) and high-yield and distress-focused hedge funds from Lipper TASS (using TASS fund classifications). Column 1 of Table 8 shows that the aggregated AUM (averaged by year) of high-yield bond mutual funds is relatively small in 2001–2002 but increases substantially (over 50%) from 2002 to 2005.

The aggregated AUM of both mutual funds and hedge funds (column 2) shows the same pattern. The availability of potential buyers' capital appears to improve over time.

We examine whether the availability of potential buyers' capital helps alleviate the impact of forced selling by insurance companies, using a price impact measure modified from Amihud (2002). For each bond, we calculate the modified price impact by dividing the absolute value of the percentage price change in the weeks following the downgrade by the total dollar trading volume of insurance companies (in millions) during the period from weeks -5 to $+5$. The percentage price change is measured as the log of the median trade price on or after week $+21$ minus the log of the median trade price during weeks -5 and $+5$. Essentially, the modified price impact measures the deviation from fundamental values (percentage price recovery) per \$1 million of insurance companies' trades.

Fig. 4 shows visually that the price impact is largest for bonds downgraded in 2001–2002, precisely over the time period when the capital held by high-yield bond mutual funds and hedge funds is comparatively low. As these buyers' capital grows from 2003 to 2005, the price impact

Table 7

Net trading volume and impairment of insurance companies.

This table relates insurance companies' net trading volume of the downgraded bonds to the impairment ratio of the insurance industry. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. Impairment ratio is the number of insurance companies becoming impaired in a given year measured as a percentage of the total number of insurance companies. Impairment ratio is reported both for each type of insurer and as an asset-weighted average of the overall industry. For each bond, the net volume is calculated as the buy volume minus the sell volume during event weeks -5 and $+5$, reported as a proportion of the bond's issue size. In the last three columns, the percentage net volume is regressed on year dummies, bond characteristic controls, and the bond-level average selling probability, respectively (based on specifications Life (1) and Property (1) in Table 4). Bond characteristic controls include logged bond age at the time of downgrade, logged issue size, logged bond maturity at the time of downgrade, and the lower than BB+ dummy. All variables, other than the year dummies, are demeaned. Robust standard errors are in parentheses. *, **, And *** refer to statistical significance at 10%, 5%, and 1% levels, respectively.

Downgrade year	(1) Life & health impairment %	(2) Property & casualty impairment %	(3) Asset-weighted impairment %	(4) Net volume as proportion of issue size	(5) Net volume as proportion of issue size	(6) Net volume as proportion of issue size
2001	0.45	1.60	0.77	-0.0274*** (0.0035)	-0.0270*** (0.0035)	-0.0245*** (0.0034)
2002	0.45	1.50	0.74	-0.0283*** (0.0029)	-0.0296*** (0.0031)	-0.0290*** (0.0031)
2003	0.20	1.15	0.47	0.0106*** (0.0035)	0.0093** (0.0038)	0.0041 (0.0047)
2004	0.30	0.60	0.39	0.0046 (0.0048)	0.0025 (0.0050)	-0.0053 (0.0067)
2005	0.60	0.50	0.57	-0.0309*** (0.0036)	-0.0343*** (0.0047)	-0.0307*** (0.0051)
Average selling probability						-0.1365** (0.0687)
Controlled for bond characteristics?				No	Yes	Yes
Observations	5	5	5	795	795	795
Adjusted R-squared				0.083	0.087	0.091

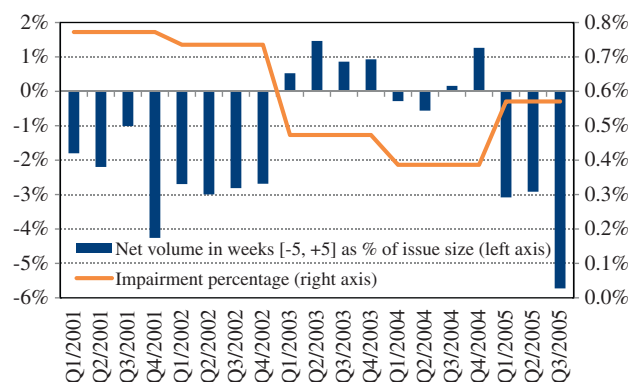


Fig. 3. Net volume and insurance industry's impairment percentage. This figure plots the time-series of insurance industry's impairment percentage (right axis) and net trading volume of downgraded bonds (left axis). Impairment percentage is the number of insurance companies becoming impaired in a given year measured as a percentage of the total number of insurance companies. Plotted is the asset value-weighted average of the life and property sectors' impairment percentages. The net volume is measured during event weeks -5 and $+5$, scaled by the issue size of each bond, and then averaged across all bonds downgraded in each quarter.

steadily declines. To further confirm this visual pattern, we regress our measure of price impact on downgrade year dummies (columns 3–5 in Table 8) and directly on logged buyers' capital in the downgrade month (column 6), controlling for market movements and (demeaned) bond characteristics. The price impact of insurance companies' selling is twice as large in 2001–2002 (ranging from 1.1% to 1.9%) as in 2003–2004 (0.6% to 0.9%). The price impact for 2005 is economically very small and statistically insignificant.

Column 6 shows that the price impact of insurance companies' selling declines by approximately 1.1% as the potential buyers' capital doubles (as happened from 2002 to 2005). Overall, the results in Table 8 and Fig. 4 are consistent with the conjecture that buyers of downgraded bonds can help alleviate the impact of forced selling by insurance companies.

Based on the above analyses, we argue that the years 2001–2002 (2003–2004) are the most (least) likely to

Table 8

Price impact and potential buyers' capital.

This table relates price impact of insurance companies' selling of the downgraded bonds to potential buyers' capital. The downgrade is from investment to non-investment grades. Modified from Amihud (2002), price impact measure is calculated as the absolute value of price reversals divided by the total dollar trading volume of insurance companies (in millions) during the period that spans from event weeks -5 to $+5$. For each bond, the price reversal is measured as the log of the median trade price on or after event week $+21$ minus the log of the median trade price during event weeks -5 and $+5$. Potential buyers' capital is measured as the assets under management (\$ millions) of high-yield bond mutual funds from the CRSP Mutual Fund Database (based on Lipper fund objective codes) and high-yield and distressed-focused hedge funds from TASS (based on TASS classifications). The monthly potential buyers' capitals are averaged and reported by year. In the last four columns, the price impact is regressed on year dummies or directly on (demeaned) log of all potential buyers' capital, market movement controls, and (demeaned) bond characteristic controls. Market movement controls are measured over the same period of time as is the price impact and include stock market return, issuer's stock return, duration-adjusted return on 10-year Treasury note, duration-adjusted return on investment-grade corporate bond spread, duration-adjusted return on high-yield corporate bond spread, and the interactions of these variables with the lower than BB+ dummy. Bond characteristic controls include logged bond age at the time of downgrade, logged issue size, logged bond maturity at the time of downgrade, and the lower than BB+ dummy. Robust standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively.

Downgrade year	(1) High-yield bond mutual funds' capital	(2) High-yield bond mutual funds' and hedge funds' capital	(3) Price impact per \$1 million	(4) Price impact per \$1 million	(5) Price impact per \$1 million	(6) Price impact per \$1 million
2001	160,390	201,760	0.0188*** (0.0025)	0.0176*** (0.0026)	0.0131*** (0.0027)	
2002	163,530	217,489	0.0188*** (0.0020)	0.0133*** (0.0028)	0.0111*** (0.0029)	
2003	230,976	305,795	0.0089*** (0.0021)	0.0062** (0.0030)	0.0062** (0.0031)	
2004	260,684	385,994	0.0082*** (0.0014)	0.0083*** (0.0028)	0.0064** (0.0027)	
2005	254,512	411,685	0.0037*** (0.0006)	0.0032** (0.0014)	0.0009 (0.0019)	
ln(Total potential buyers' capital)						−0.0147*** (0.0046)
Constant						0.0079*** (0.0012)
Controlled for market movement?			No	Yes	Yes	Yes
Controlled for bond characteristics?			No	No	Yes	Yes
Observations	55	55	495	404	404	404
Adjusted R-squared			0.066	0.105	0.180	0.185

result in fire sale prices of downgraded issues. Year 2005 is ambiguous since, although life insurance companies are under stress and sell downgraded bonds in large volume, potential buyers' funds appear sufficient to absorb the sales. Consistent with the prediction of Shleifer and Vishny (1992), Fig. 5 shows that MCARs are significantly more negative for bonds downgraded in 2001–2002 than for bonds downgraded in 2003–2005. While we acknowledge the limited number of temporal observations given the span of our data, this result nonetheless suggests that the severity of price declines around downgrades is indeed significantly larger when fire sales are more likely to happen—in the face of broad industry distress and limited buyers' availability.

4.3. Multivariate analysis of price reversals

The evidence in Table 6 and Fig. 1 is suggestive of the relationship between regulatory constraints and fire sales.²⁷ We take as evidence of fire sales the price deviation from fundamental values, exhibited by the

difference in price declines and subsequent reversals between the bonds with high average selling probability (i.e., held mostly by relatively constrained insurers) and those with low average selling probability. The limited number of downgraded bonds restricts our ability to trace the full path of prices from weeks -20 to $+35$; hence, we split the sample into two groups and compare their MCARs. In this section, we perform a more powerful multivariate analysis to identify the direct effects of regulatory constraints on the fire sale price deviation from fundamental value. Since transaction prices of downgraded bonds fully recover after 30 or so weeks, we can reasonably use as a measure of the fire sale price deviation the price reversals from the fire sales period to the full recovery period. This way, we retain a much larger set of bonds for our analysis.

Theoretically, price reversals are the change in transaction prices from the fire sales period to a later period in which prices have reached fundamental values. The price reversals, therefore, measure the temporary price discount due to fire sales. Based on the patterns of MCARs in Figs. 2 and 5, we define the fire sales period as event weeks -5 to $+5$. We count each trade in this period as one observation so that bonds that trade multiple times in this period will carry more weight. The actual trade price $P_{i,t}$ is used. Given the observed pattern of price recovery, we choose the period from week $+21$ onwards as the period T in which price reversals have completed. For

²⁷ The observed magnitude of the price reversal appears too large to be explained by information asymmetry. For example, Cai, Helwege, and Wargha (2007) find that underpricing on speculative-grade debt Initial Public Offerings (IPOs) averages only about 47 basis points. Edwards, Harris, and Piwowar (2007) find that round-trip transaction costs for speculative-grade bonds average less than 50 basis points for trades larger than \$1 million in size.

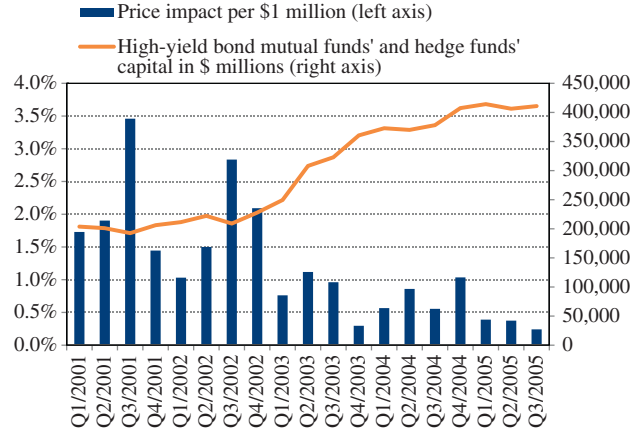


Fig. 4. Price impact and potential buyers' capital. This figure plots the time-series of potential buyers' capital (right axis) and price impact of insurance companies' selling of downgraded bonds (left axis). Potential buyers' capital is measured as the assets under management (\$ millions) of high-yield bond mutual funds from the CRSP Mutual Fund Database (based on Lipper fund objective codes) and high-yield and distressed-focused hedge funds from TASS (based on TASS classifications). Modified from Amihud (2002), price impact measure is calculated as the absolute value of price reversals divided by the total trading volume of insurance companies (in \$ millions) during event weeks -5 and $+5$. The price reversal for each bond is measured as the log of the median trade price on or after event week $+21$ minus the log of the median trade price during event weeks -5 and $+5$. Plotted is the average of price impact across all bonds downgraded in each quarter.

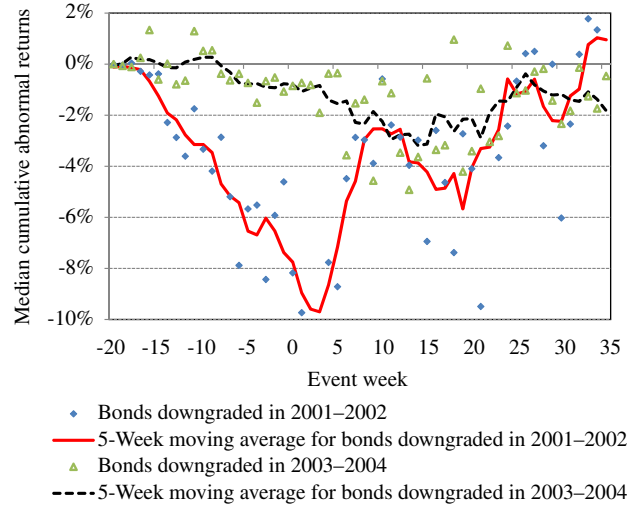


Fig. 5. Median cumulative abnormal returns by year of downgrade. This figure plots the median cumulative abnormal returns (MCARs) for bonds downgraded in 2001–2002 vs. 2003–2004. For each bond, cumulative abnormal returns (CARs) are calculated by the model-adjustment methodology, as described in Table 6 and Fig. 2. The calculation uses only actual trade prices on bonds that trade at least once before week -20 , at least once between weeks -20 and 0 , at least once between weeks 0 and $+35$, and at least once after week $+35$, i.e., Sample 5.1 in Table 1. For each event week, MCARs are measured as the medians of CARs across all transactions and all bonds downgraded in 2001–2002 and in 2003–2004.

each bond, we calculate the benchmark price $\bar{P}_{i,T}$ as the median price of all the trades in this period. Thus, only bonds that trade at least once during event weeks -5 to $+5$ and at least once in or after event week $+21$ enter our analysis. The price reversal for bond i based on fire sale trade t is calculated as

$$R_{i,t,T} = \ln(\bar{P}_{i,T}/P_{i,t}), \quad \forall t \in [\text{weeks } -5, +5] \text{ and } [\text{weeks } +21, \text{ end of sample}].$$

To test our hypothesis that the fire sales of downgraded bonds are driven by regulatory pressure, we regress price reversals on a proxy of bond-level regulatory constraints.

We control for changes in bond and stock market conditions, issuer-specific information, transaction costs, and other bond characteristics. We also allow for bonds that are downgraded to a lower rating to load differently on the market control variables, using a category dummy variable. Our model is as follows:

$$R_{i,t,T} = \beta_0 + \beta_C C_{i,t} + \beta_X X_{i,t} + \beta_D D_i + \beta_Z Z_{t,T} + \beta_{ZD} D_i Z_{t,T} + (\bar{A}_{i,T} \bar{Q}_{i,T} - A_{i,t} Q_{i,t}) + \zeta_{i,t,T}, \quad (7)$$

where $C_{i,t}$ is the proxy of regulatory constraint for bond i on date t , $X_{i,t}$ is a vector of bond static and time-varying characteristics, D_i is a dummy variable equal to one if

bond i is downgraded to a rating lower than BB+ and zero otherwise, $Z_{i,T}$ is a vector of changes in market conditions from dates t to T , $A_{i,t}$ is a percentage half spread for trade t on bond i specified as in (3), and $Q_{i,t}$ equals one (–1) for a buy (sell) trade. The upper bar denotes the trade that determines the benchmark price in period T .

We allow the bond-level regulatory constraints to enter our model both in linear fashion, e.g., average logged RBC ratio, and in a non-linear fashion, e.g., percentage of holding by insurers with RBC ratios in the bottom 25%. The latter is to address the concern, discussed earlier, that only insurance companies close to the regulatory-action threshold likely face forced selling.²⁸ In addition to the bond-level measures of regulatory constraints (based on the constraints faced by holders of the bonds), we also use logged potential buyers' capital as a measure of the market's ability to absorb the sales of downgraded bonds. We estimate the parameters of (7) by Ordinary Least Squares (OLS) using heteroskedasticity-adjusted standard errors.²⁹ Table 9 reports the results.

The results in the first five columns of Table 9 Panel A, using different measures of bond-level regulatory constraints, show unanimously that bonds whose holders are more constrained exhibit significantly larger price reversals. For example, the third column shows that the lower the average RBC ratio of insurance companies holding the downgraded bond (relatively more constrained), the larger the price reversal. If we shock (the log of) the average RBC ratio from the 25th to the 75th percentiles (becoming less constrained), the price reversal will decrease by 1.2% (mean reversal is 7.2%). Focusing only on insurers close to the regulatory threshold, the fourth column examines the effect of bond holdings by companies with RBC ratios in the bottom 25% of the sample. The results show that the more widely the bond is held by these constrained insurers, the larger the price reversals. Increasing the holding by these constrained insurers from the 25th to the 75th percentiles increases the price reversal by 1.5%.

In the last two columns, we investigate the influence of (the log of) potential buyers' capital (for the downgrade month of each bond) on price reversals, controlling for bond-level regulatory constraints. The last two columns of Table 9 show that, consistent with the fire sales hypothesis, price reversals are significantly smaller when buyers' capital is more available than when it is scarce.

Finally, given that both the price reversals and our right-hand-side variables are highly skewed, we also estimate Eq. (7) by quantile regression (simultaneously

for the 0.25, 0.50, and 0.75 quantiles). The quantile regression estimates in Table 9 Panel B largely confirm our OLS results.

5. Alternative explanations and robustness checks

Our findings are consistent with the fire sales hypothesis: (i) constrained insurance companies are more likely to sell the downgraded bonds immediately after the downgrade, (ii) while price effects are limited, on average, bonds held widely by constrained companies experience a significant price discount from fundamental values, and finally, (iii) bonds downgraded during periods in which the insurance industry is under stress and the potential buyers' capital is relatively scarce experience a significant price discount to fundamental values. In this section, we demonstrate the robustness of these findings to a variety of methods and controls, and explore some alternative hypotheses.

5.1. Information contained in the downgrade

Thus far, we take the 7% difference between MCARs of the bonds with above-median average selling probability (held mostly by constrained insurers) and those of the bonds with below-median average selling probability (held mostly by unconstrained insurers) as an indication of the price effects of fire sales. However, these effects can be explained by two possible hypotheses: fire sales or the information released by the credit rating downgrade. Arguably, the latter is unlikely since we control for issuer-specific information using issuer stock returns in our benchmark model. Also, most importantly, we focus only on the *difference* between two groups of bonds delineated by the constraints the bond holders face. These constraints are likely to be orthogonal to the information content in each downgrade.

Still, it is possible that the downgrade's information amplifies the effects of fire sales. To isolate the fire sales effects, we follow Ambrose, Cai, and Helwege (2009) and use a sample of bonds that have insignificant issuer stock returns around the downgrade. Ambrose, Cai, and Helwege (2009) argue that this type of sample is the most conducive to investigate sales that are "highly likely to represent regulatory pressure rather than information-motivated trading." Of the 584 bonds that we identify as having insignificant stock price reaction, 237 bonds meet our requirements for abnormal return studies (Sample 6.1 in Table 1). As in the earlier analyses, these bonds are divided into the more constrained and less constrained groups by (i) bond-level average selling probability, and (ii) years of downgrade (2001–2002 vs. 2003–2004). The MCARs are then calculated as the median of CARs for each constraint group in each event period. Table 10 reports the results.

The results in both panels of Table 10 show that the price response still depends on the regulatory constraints faced by the insurers holding the downgraded bonds and the availability of buyers' capital. First, the MCARs for the zero-stock-return bonds that are classified as less constrained are very small and mostly statistically insignificant. This result

²⁸ Bessembinder, Kahle, Maxwell, and Xu, 2009 suggest that analysis of abnormal returns should be performed at the issuer level to avoid being dominated by some large issuers with several bond issues. While we report the results using bond-level data, we also run a robustness check by using only one random bond from each issuer. The results (unreported for brevity) do not change.

²⁹ We also obtain the same results using Iterated Weighted Least Squares (IWLS) in which we place higher weight on more precise observations, as proxied by the shorter time period between the trades during weeks –5 and +5 and the benchmark trades (in or after week +21).

Table 9

Multivariate regression of price reversals.

This table reports coefficient estimates for the regression of price reversals on measures of regulatory constraint of an average insurance company holding the downgraded bonds. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. For each transaction executed during the period from event weeks -5 to $+5$, price reversal (the dependent variable) is measured as the logged benchmark price minus the logged transaction price. The benchmark price for each bond is measured as the median price of all transactions that are executed in or after event week $+21$. All other change and return variables are measured in the same fashion. Age and years to maturity are measured at the time of downgrade. Logged potential buyers' capital (\$ millions) is for the month of downgrade. Panel A reports OLS estimates for seven models with different measures of regulatory constraints, as listed in the column headings. White's robust standard errors are in parentheses. Panel B reports quantile regression estimates (0.25, 0.50, and 0.75) using two different measures of regulatory constraints, one using % holding by property insurance companies and the other using average holders' selling probability. Bootstrapped standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively. Variable definitions are in Appendix A and B.

	Bond-level regulatory constraint						
	% Holding by property insurers	Average selling probability	Average ln(RBC ratio)	% Holding by low RBC ratio	Average ln(RACR1)	% Holding by property insurers	Average selling probability
<i>Panel A: OLS regressions</i>							
Bond-level regulatory constraint	0.0018*** (0.0004)	0.3502*** (0.0491)	-0.1208*** (0.0277)	0.0008*** (0.0169)	-0.1341*** (0.0222)	0.0011*** (0.0004)	0.3138*** (0.0467)
ln(Buyers' capital)						-0.2129*** (0.0082)	-0.2093*** (0.0084)
<i>Insurance company controls</i>							
Average ln(capital & surplus)		-0.0425*** (0.0041)	-0.0426*** (0.0041)	-0.0404*** (0.0042)	-0.0427*** (0.0042)		-0.0344*** (0.0040)
Average risky asset holding		0.0034*** (0.0009)	0.0027*** (0.0009)	0.0029*** (0.0009)	0.0030*** (0.0009)		-0.0005 (0.0008)
<i>Market movement controls</i>							
Stock market return	0.0429*** (0.0156)	0.0711*** (0.0157)	0.0631*** (0.0154)	0.0696*** (0.0154)	0.0678*** (0.0156)	0.0977*** (0.0146)	0.1198*** (0.0146)
[Stock market return] $\times$ [Lower than BB+ dummy]	-0.1097*** (0.0275)	-0.1232*** (0.0277)	-0.1100*** (0.0275)	-0.1122*** (0.0274)	-0.1238*** (0.0280)	-0.0739*** (0.0251)	-0.0927*** (0.0251)
Issuer's stock return	0.0706*** (0.0050)	0.0781*** (0.0050)	0.0746*** (0.0049)	0.0742*** (0.0049)	0.0709*** (0.0049)	0.0788*** (0.0047)	0.0840*** (0.0047)
[Issuer's stock return] $\times$ [Lower than BB+ dummy]	0.0437*** (0.0096)	0.0290*** (0.0096)	0.0346*** (0.0096)	0.0312*** (0.0097)	0.0354*** (0.0096)	0.0106 (0.0092)	-0.0007 (0.0091)
Treasury bond return	0.3537*** (0.0609)	0.2924*** (0.0585)	0.2929*** (0.0571)	0.2895*** (0.0580)	0.2914*** (0.0575)	0.2923*** (0.0388)	0.2469*** (0.0382)
[Treasury bond return] $\times$ [Lower than BB+ dummy]	0.0453 (0.0874)	0.0463 (0.0866)	0.0350 (0.0854)	0.0450 (0.0860)	0.0265 (0.0860)	-0.1483*** (0.0709)	-0.1407*** (0.0716)
Investment-grade spread return	-0.1152 (0.1077)	-0.0900 (0.1036)	-0.0246 (0.1020)	-0.0628 (0.1018)	-0.0516 (0.1030)	-0.1593*** (0.0797)	-0.1437* (0.0772)
[Investment-grade spread return] $\times$ [Lower than BB+ dummy]	0.2958* (0.1643)	0.2722* (0.1641)	0.2552 (0.1633)	0.2446 (0.1636)	0.2659 (0.1639)	0.3694** (0.1482)	0.3063** (0.1491)
High-yield spread return	0.1139*** (0.0252)	0.1063*** (0.0243)	0.0880*** (0.0240)	0.0962*** (0.0240)	0.0944*** (0.0240)	0.0415** (0.0177)	0.0374** (0.0174)
[High-yield spread return] $\times$ [Lower than BB+ dummy]	0.0233 (0.0404)	0.0301 (0.0403)	0.0270 (0.0399)	0.0385 (0.0402)	0.0303 (0.0400)	0.0239 (0.0358)	0.0400 (0.0359)
<i>Transaction cost controls</i>							
Δ Trade Direction	0.0144 (0.0111)	0.0173 (0.0107)	0.0180* (0.0108)	0.0196* (0.0108)	0.0151 (0.0107)	0.0358*** (0.0105)	0.0326*** (0.0101)
Δ (Trade direction $\times$ ln(Trade size))	0.0002 (0.0008)	-0.0002 (0.0008)	-0.0001 (0.0008)	-0.0003 (0.0008)	0.0000 (0.0008)	-0.0014* (0.0008)	-0.0013* (0.0007)
<i>Bond controls</i>							
Lower than BB+ dummy	0.0568*** (0.0045)	0.0556*** (0.0046)	0.0521*** (0.0046)	0.0501*** (0.0046)	0.0523*** (0.0045)	0.0513*** (0.0043)	0.0510*** (0.0044)
ln(Age in years)	-0.0207*** (0.0030)	-0.0237*** (0.0036)	-0.0263*** (0.0037)	-0.0298*** (0.0034)	-0.0323*** (0.0033)	0.0174*** (0.0031)	0.0131*** (0.0035)
ln(Issue size)	0.0038* (0.0020)	-0.0160*** (0.0024)	-0.0060*** (0.0022)	-0.0061*** (0.0021)	-0.0081*** (0.0021)	0.0299*** (0.0022)	0.0131*** (0.0026)
ln(Years to maturity)	-0.0211*** (0.0033)	-0.0155*** (0.0033)	-0.0133*** (0.0033)	-0.0180*** (0.0033)	-0.0153*** (0.0033)	-0.0085*** (0.0031)	-0.0022 (0.0030)
Constant	0.0303 (0.0292)	1.0243*** (0.0952)	1.1779*** (0.1025)	0.8912*** (0.0982)	1.1144*** (0.0965)	2.2835*** (0.0915)	3.1105*** (0.1238)
Observations	4,067	4,067	4,067	4,067	4,067	4,067	4,067
Adjusted R-squared	0.405	0.428	0.425	0.426	0.427	0.488	0.504

	% Holding by property insurers			Average selling probability		
	25%	50%	75%	25%	50%	75%
<i>Panel B: Quantile regressions</i>						
Bond-level regulatory constraint	0.0012*** (0.0004)	0.0015*** (0.0004)	0.0005 (0.0007)	0.2496*** (0.0850)	0.4265*** (0.0638)	0.5037*** (0.0604)
ln(Buyers' capital)	−0.1553*** (0.0093)	−0.1894*** (0.0103)	−0.2479*** (0.0121)	−0.1639*** (0.0114)	−0.1800*** (0.0109)	−0.2352*** (0.0137)
Insurance company controls	Yes	Yes	Yes	Yes	Yes	Yes
Market movement controls	Yes	Yes	Yes	Yes	Yes	Yes
Transaction cost controls	Yes	Yes	Yes	Yes	Yes	Yes
Bond controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,067	4,067	4,067	4,067	4,067	4,067
Pseudo R-squared	0.247	0.301	0.387	0.261	0.323	0.404

Table 10

Cumulative abnormal returns for downgraded bonds with zero issuers' stock returns.

This table reports the median cumulative abnormal returns (MCARs) for the sample of downgraded bonds whose issuers' stock returns upon the downgrade are not significantly different from zero. Following [Ambrose, Cai, and Helwege \(2009\)](#), the zero-issuer-stock-return sample is identified by comparing the average daily stock return over the three-day event window (days −1 to +1) with the average historical stock return over 250 days (days −280 to −31). Using 5% significance level, the sample contains 584 bonds, but only 237 bonds meet the screening criteria and are used in the calculation of MCARs (Sample 6.1 in [Table 1](#)). The methods for (a) calculating abnormal returns (ARs), (b) accumulating ARs over event time to obtain cumulative abnormal returns (CARs) for each bond, and (c) aggregating CARs across bonds to obtain MCARs are as described in [Table 6](#). The included bonds are divided into the more constrained and less constrained groups by (i) average selling probability (Panel A) calculated as described in [Table 6](#) and (ii) years of downgrade (Panel B). Based on (i), the bonds with average selling probability above (below) the total sample median (Sample 4 in [Table 1](#)) are considered more (less) constrained. Based on (ii), the bonds downgraded in 2001–2002 (2003–2004) are considered more (less) constrained. The *t*-statistics are for the tests of (a) null hypothesis that MCAR equals zero and (b) null hypothesis that MCARs for the more and less constrained groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds in the more constrained group).

5-Week period	More constrained group		Less constrained group		Difference	
	(1) MCAR	H0: MCAR=0 t-statistic	(2) MCAR	H0: MCAR=0 t-statistic	(1)−(2) MCAR	H0: (1)=(2) t-statistic
<i>Panel A: By average selling probability</i>						
[−20, −16]	0.000	(0.00)	0.000	(0.00)	0.000	(0.00)
[−15, −11]	−1.262	(−2.52)	−0.389	(−1.79)	−0.874	(−1.88)
[−10, −6]	−2.442	(−1.67)	−0.351	(−1.00)	−2.090	(−1.18)
[−5, −1]	−3.445	(−4.23)	−0.833	(−2.21)	−2.613	(−2.65)
Week 0	−4.500	(−2.59)	−0.483	(−0.53)	−4.017	(−1.66)
[1, 5]	−3.960	(−2.66)	−0.162	(−0.50)	−3.798	(−2.66)
[6, 10]	−4.221	(−3.30)	−0.421	(−1.25)	−3.800	(−2.58)
[11, 15]	−2.567	(−2.16)	−1.204	(−1.98)	−1.363	(−1.03)
[16, 20]	−3.925	(−1.53)	−2.440	(−3.17)	−1.485	(−0.55)
[21, 25]	−1.815	(−1.08)	−1.064	(−1.04)	−0.752	(−0.61)
[26, 30]	−3.428	(−1.86)	0.275	(0.32)	−3.703	(−1.71)
[31, 35]	−0.523	(−0.27)	0.325	(0.38)	−0.849	(−0.42)
<i>Panel B: By downgrade year</i>						
[−20, −16]	0.019	(0.09)	0.000	(0.00)	0.019	(0.12)
[−15, −11]	−1.262	(−2.63)	−0.137	(−0.60)	−1.125	(−2.31)
[−10, −6]	−3.417	(−2.68)	0.359	(0.87)	−3.776	(−3.25)
[−5, −1]	−3.893	(−4.62)	−0.430	(−0.82)	−3.463	(−3.82)
Week 0	−2.637	(−1.09)	−0.091	(−0.08)	−2.546	(−1.21)
[1, 5]	−3.881	(−2.59)	−0.063	(−0.14)	−3.817	(−2.58)
[6, 10]	−2.343	(−1.00)	−1.007	(−1.64)	−1.336	(−0.42)
[11, 15]	−3.149	(−3.41)	−1.145	(−2.09)	−2.004	(−1.95)
[16, 20]	−4.218	(−3.44)	−1.882	(−1.32)	−2.336	(−1.25)
[21, 25]	−3.692	(−1.67)	−0.763	(−0.77)	−2.928	(−1.12)
[26, 30]	−1.282	(−0.94)	−0.080	(−0.12)	−1.202	(−0.81)
[31, 35]	0.346	(0.24)	−0.885	(−1.11)	1.231	(0.70)

holds whether we use average selling probability or years of downgrade as the measure of constraint. We interpret this result as consistent with the findings of [Ambrose, Cai, and](#)

[Helwege \(2009\)](#) in that they find generally limited price effects. In our case, the MCARs for the zero-stock-return bonds that are classified as more constrained are around

3–4% lower (more negative) than those for the less constrained bonds. Since the downgrades of these zero-stock-return bonds contain minimal new information, we interpret this 3–4% difference in MCARs as the pure regulatory-induced fire sales effects.

5.2. Large selling volume

One obvious alternative explanation of our results is the pure price pressure hypothesis offered by Scholes (1972). Transaction prices diverge from fundamental values because of an uninformed shock caused by either excess demand or excess supply. The evidence shows that the deviation of transaction prices from fundamental values can last from a few hours (Gemmill, 1996; Ellul, 2006) and up to a few days (Harris and Gurel, 1986; Kraus and Stoll, 1972; Scholes, 1972; Shleifer, 1986).

We investigate the pure price pressure explanation by reproducing the MCARs around large sell trades that are unrelated to obvious important events such as a downgrade. Specifically, we define as an event for each bond the large sell trade comparable in size to the selling volume around the downgrade. We find no significant price declines or reversals, suggesting that the simple liquidity story does not explain the price patterns we observe for corporate bond downgrades. Rather, this finding supports the view that the collective need of insurance companies to sell immediately and the lack of unconstrained buyers are of first-order importance.

5.3. Do insurance companies try to avoid fire sales?

To reduce their portfolio risk and the associated implications for regulatory capital, insurance companies can sell either the downgraded bonds or some other risky bonds they hold. The latter should be preferred given that the downgraded bonds are likely to trade at fire sale prices. We examine trading activities in *other* bonds held by the same insurance companies that hold the downgraded bonds. We find that insurance companies holding the downgraded bonds seem to sell other speculative-grade bonds more intensely than other companies that hold the same bonds. However, the magnitude of the difference is rather small ranging from about \$30,000 per bond per company per week in the event week to about \$6,000 during weeks +6 and +10.

Finally, the fact that affected insurance companies sell other speculative-grade bonds upon the downgrade of a nominally unrelated bond raises an important question: are there spillover effects for prices from the downgraded bonds to other speculative-grade bonds? We find that other speculative-grade bonds held by affected insurance companies also experience price declines that take place right after the downgrade events and price reversals that occur slowly over the next several weeks. However, the magnitude of the spillover effect is economically quite small, probably because the *collective* divestment following a downgrade is absent. These results are available upon request.

5.4. Alternative methodologies for calculating abnormal returns

Finally, to investigate the robustness of our results, we also calculate MCARs using two other methods (see Brown and Warner, 1985; Bessembinder, Kahle, Maxwell, and Xu, 2009): (i) mean adjustment, and (ii) bond index adjustment. The mean adjusted abnormal returns equal the actual excess bond returns minus the mean excess return during the estimation period. Excess returns are defined as the difference between the corporate bond returns and the returns on Treasury bonds in the same maturity group. We use the Barclays Treasury Bond Index (price) for maturity groups 1–3 years, 3–5 years, 5–7 years, 7–10 years, 10–20 years, and > 20 years as the benchmarks.³⁰ The bond-index adjusted abnormal returns are calculated as the actual bond returns minus the rating- and maturity group-matched Barclays Bond Index (price) returns. We use the Barclays Corporate Bond Index (price) for maturity groups less than 15 years and greater than 15 years and for ratings A, BBB, BB, B, and CCC.

Note that both mean adjustment and bond-index adjustment do not account for issuer-specific information. To mitigate a potential overestimation of fire sales effects due to information, we apply these methodologies only to the zero-issuer-stock-return sample discussed in Section 5.1. Table 11 reports MCARs from weeks –10 to +10 by constraint groups. Panels A and C are based on bond-index adjustment method. Panels B and D are based on mean adjustment method. The fire sales effects we showed earlier, heterogeneity in the price effects across groups of bonds separated by relative constraints, remain robust: the MCARs for the zero-stock-return bonds that are classified as relatively more constrained (high average selling probability or downgraded in 2001–2002) are around 2–6% lower (more negative) than those for the less constrained bonds.

6. Conclusions

This paper investigates fire sales of corporate bonds by insurance companies. These companies operate under regulations that constrain their risk-taking capacity. An insurance company that faces binding regulatory constraints often has to sell some of its risky assets. Since insurance companies as a group hold over a third of outstanding investment-grade corporate bonds, any event that forces them to immediately and collectively sell the same bonds can induce a fire sale.

We study the trading behavior of regulated insurance companies and the associated bond price patterns when investment-grade corporate bonds are downgraded to a speculative grade (a rating of BB+ or below). To empirically test the fire sales hypothesis, we construct a data set of all the corporate bond downgrades and individual insurance companies' positions and transactions from 2001 to 2005. The heterogeneity afforded by our use of

³⁰ Barclays Bond Indexes are formerly known as Lehman Brothers Indexes.

Table 11

Cumulative abnormal returns by alternative methodologies.

This table reports the median cumulative abnormal returns (MCARs), calculated by two alternative methodologies of measuring abnormal returns: (i) bond-index adjustment (Panels A and C), and (ii) mean adjustment (Panels B and D). *Bond-index adjusted* abnormal returns are calculated as actual bond returns minus rating- and maturity group-matched Barclays bond (price) index return. *Mean adjusted* abnormal returns are calculated as actual bond excess returns minus the mean excess return before event week -20 (scaled by number of days between trades). Excess returns are defined as the difference between corporate bond returns and returns on Treasury bonds in the same maturity group. The calculation uses only the zero-issuer-stock-return sample, Sample 6.1 in Table 1. The methods for (a) accumulating abnormal returns over event time to obtain cumulative abnormal returns (CARs) for each bond, and (b) aggregating CARs across bonds to obtain MCARs are as described in Table 6. The included bonds are divided into the more constrained and less constrained groups by (i) average selling probability (Panels A and B) calculated as described in Table 6 and (ii) years of downgrade (Panels C and D). Based on (i), the bonds with average selling probability above (below) the total sample median (Sample 4 in Table 1) are considered more (less) constrained. Based on (ii), the bonds downgraded in 20012002 (20032004) are considered more (less) constrained. The *t*-statistics are for the tests of (a) null hypothesis that MCAR equals zero and (b) null hypothesis that MCARs for the more and less constrained groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds in the more constrained group).

5-Week period	More constrained group		Less constrained group		Difference	
	(1) MCAR	H0: MCAR=0 t-statistic	(2) MCAR	H0: MCAR=0 t-statistic	(1) - (2) MCAR	H0: (1)=(2) t-statistic
<i>Panel A: Index-adjusted MCARs by average selling probability</i>						
[−10, −6]	−3.260	(−3.70)	−0.039	(−0.08)	−3.221	(−2.73)
[−5, −1]	−4.134	(−3.96)	−0.271	(−0.37)	−3.862	(−3.09)
Week 0	−3.762	(−1.88)	−0.767	(−1.25)	−2.995	(−1.62)
[1, 5]	−3.979	(−2.93)	−0.876	(−1.49)	−3.103	(−2.07)
[6, 10]	−3.307	(−2.84)	−1.975	(−3.20)	−1.331	(−1.12)
<i>Panel B: Mean-adjusted MCARs by average selling probability</i>						
[−10, −6]	−3.015	(−2.95)	−1.196	(−1.62)	−1.818	(−1.36)
[−5, −1]	−4.868	(−5.09)	−2.332	(−2.74)	−2.536	(−1.87)
Week 0	−4.291	(−2.16)	−2.313	(−3.66)	−1.978	(−0.82)
[1, 5]	−5.205	(−4.71)	−1.583	(−2.47)	−3.622	(−2.93)
[6, 10]	−5.289	(−2.99)	−1.256	(−2.04)	−4.033	(−1.89)
<i>Panel C: Index-adjusted MCARs by downgrade year</i>						
[−10, −6]	−5.720	(−5.60)	−0.285	(−0.53)	−5.435	(−3.85)
[−5, −1]	−5.398	(−2.94)	−0.394	(−0.97)	−5.004	(−2.83)
Week 0	−4.426	(−1.77)	−0.850	(−0.82)	−3.576	(−1.02)
[1, 5]	−5.724	(−3.14)	−0.876	(−1.85)	−4.848	(−2.38)
[6, 10]	−3.784	(−2.15)	−2.434	(−3.89)	−1.350	(−0.73)
<i>Panel D: Mean-adjusted MCARs by downgrade year</i>						
[−10, −6]	−6.938	(−5.15)	0.658	(1.46)	−7.596	(−4.86)
[−5, −1]	−7.168	(−4.17)	0.584	(0.66)	−7.752	(−4.10)
Week 0	−4.927	(−2.11)	0.875	(0.93)	−5.801	(−3.21)
[1, 5]	−5.695	(−6.00)	−0.150	(−0.28)	−5.545	(−4.45)
[6, 10]	−3.382	(−1.82)	−0.658	(−1.51)	−2.724	(−1.39)

individual insurance company data is fundamental to our analysis as it permits the disentangling of forced selling from other discretionary trades.

In support of the hypothesized role for regulation, we find that regulatory-constrained insurance companies with, for example, low risk-based capital ratios are more likely to sell the downgraded bonds immediately after the downgrade. Also, we show that around the downgrade, the price deviations from fundamental values are significantly larger for bonds held widely by these constrained firms. For instance, the median cumulative abnormal returns in the first five weeks after the downgrade are almost -9% for bonds subject to high probability of forced selling but less than -3% for other downgraded bonds. This difference of about 6% disappears after 30 weeks. Finally, bonds downgraded during periods in which the insurance industry is under stress and the potential buyers' capital is relatively scarce experience a significant price discount to fundamental values.

The conventional wisdom regarding corporate bond downgrades fails to reflect the critical role for the regulatory constraints faced by the individual insurance companies holding these downgraded bonds. Our tests all point to one conclusion: regulatory constraints imposed on insurance companies are important in explaining fire sales in the corporate bond market.

Appendix A. Descriptions of insurance company-specific variables

See Table A1.

Appendix B. Descriptions of other variables

See Table B1.

Table A1

Variable	Definition
Capital and surplus	The insurance company's statutory net worth (including paid-in capital/unimpaired surplus and additional funds in surplus) in millions of dollars through the most recent year-end
Holding of non-investment-grade bonds	The percentage of investment assets invested in non-investment-grade bonds (bonds that do not hold an investment-grade rating by any qualified rating agency)
Holding of risky assets	The percentage of investment assets invested in any of the following asset classes: non-investment-grade bonds, common and preferred stocks, non-performing mortgages, real estate, and other investments. According to the Street.com and NAIC, the target capital percentages for these assets are greater than or equal to those of the least risky class of non-investment-grade bonds (BB)
Invested assets	The reported market value in millions of dollars of the insurance company's admitted assets (i.e., assets recognized and accepted by the state insurance regulators in determining solvency of the company) that are invested in the capital and money markets through the most recent year-end
NAIC risk-based capital ratio (RBC ratio)	The ratio of total adjusted capital (capital, surplus, and applicable valuation reserves) to NAIC risk-based capital (RBC). RBC is the <i>minimum</i> amount of capital that the insurance company must maintain based on the inherent risks in its operations. RBC is calculated based on the NAIC's formula which reflects its assessment of risks of different asset classes and businesses. For example, a company with RBC ratio of 1.0 has capital equal to its RBC. Insurance companies with higher RBC ratios are considered better capitalized. Insurance companies with RBC ratio below 2.0 are subject to supervisory interventions. The levels of supervisory actions (ranging from company action plus reporting to the regulator to the regulator taking control of the company) depend on the level of RBC ratio. However, companies with RBC ratios between 2.0 and 3.0 (viewed in combination with the combined ratio) may still be required to take action to improve capitalization if they are licensed in the regulatory jurisdictions that utilize the so-called "Trend Tests"
Risk-adjusted capital ratio 1 (RACR1) ^a	The ratio of capital base to target capital (similar in spirit to the NAIC's RBC ratio). Both the capital base and the target capital are determined based on the Street.com's formula. Capital base measures the insurance company's resources that can be used to cover losses. These resources include capital, surplus, applicable valuation reserves, and other applicable provisions. Target capital is the amount of capital resources that the Street.com feels the insurance company would need to cover potential losses. The target capital for a moderate loss scenario is used in calculating RACR1. The formula expresses the target capital as a percentage of the investment risk and business risk exposures. Generally, the Street.com's target capital percentages are higher (more stringent) than the NAIC RBC percentages. The critical level that indicates serious undercapitalization is 1.0

^a The Street.com may not evaluate some insurance companies for one or more of the following reasons: (i) total assets are less than \$1 million, (ii) premium income for the current year is less than \$100,000, (iii) the company functions almost exclusively as a holding company rather than as an underwriter, or (iv) the Street.com does not have enough information to reliably evaluate the company.

Table B1

Variable	Definition
Age in years	Time from issuance to the downgrade event or the trade in question depending on specification, measured in years
Average "X"	Bond-level variable calculated by averaging the insurance company-level variable "X" across all insurance companies holding the bond at calendar quarter-end preceding the downgrade
Buyers' capital	Buyers' capital is the assets under management (\$ million) of high-yield bond mutual funds from the CRSP Mutual Fund Database (based on Lipper fund objective codes) and high-yield and distressed-focused hedge funds from TASS (based on TASS classifications). For each bond, buyers' capital is measured in the month of downgrade announcement
Lower than BB+ dummy	Dummy variable equal to one for bonds that are downgraded to a rating lower than BB+, and zero otherwise
High-yield spread return	Return on Barclays (formerly known as Lehman Brothers) High-Yield Bond Index, measured as the product of change in average credit spreads of bonds in the index (over yield of the 10-year Treasury note) from the prior trading day to the current trading day and maturity of the bond on the previous trading day
% Holding by low RBC ratio	Holding of a bond issue by insurance companies with RBC ratios in the bottom 25% (corresponding to the value of approximately 5.0) as a percentage of all insurance companies' holding of the bond
% Holding by property insurers	Percentage of the bond issue held by property insurance companies
Impairment ratio (or percentage)	Impairment ratio is the number of insurance companies becoming impaired in a given year measured as a percentage of the total number of insurance companies
Investment-grade spread return	Return on Barclays (formerly known as Lehman Brothers) Investment-Grade Bond Index, measured as the product of change in average credit spreads of bonds in the index (over yield of the 10-year Treasury note) from the previous trading day to the current trading day and maturity of the bond on the previous trading day
Issue size	Dollar offering amount of the bond
Property insurance dummy	Dummy variable equal to one for property and casualty insurance companies, and zero for life and health companies
Stock market return	Percentage change in the CRSP value-weighted total return index from the prior trading day
Δ Trade direction	Trade direction of the current trade minus trade direction of a trade in the prior trading day. Trade direction is an indicator variable, equal to one for a buy and −1 for a sell
Trade size	Par value of transaction
Treasury bond return	Return on a constant-maturity 10-year Treasury note, measured as the product of the change in yield from the previous trading day to the current trading day and maturity of the bond on the previous trading day. For variable-rate bonds, we use the maturity of one year in this calculation
Years to maturity	Maturity of the bond at the time of trade or at the time of downgrade depending on specifications, measured in years

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